

# SCX-Audited Macroeconomic Forecasting: Multi-Model Consensus under Structural Breaks

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SCX

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## Abstract

We present **SCX-Economics**, a formal framework for audited macroeconomic forecasting that integrates multiple heterogeneous forecasting models under a unified consensus mechanism. Our approach treats macro forecasting models — Dynamic Stochastic General Equilibrium (DSGE), Vector Autoregression (VAR/BVAR), agent-based models (ABM), and machine learning models (random forest, LSTM) — as *experts* whose forecasts are aggregated via a principled voting scheme (**Yajie** aggregation). The economic regime (expansion, recession, stagflation, or crisis) serves as the latent state driving model selection and regime-switching dynamics.

We establish three foundational theorems. **Theorem 1** (Multi-Model Forecast Error Detection) provides a concentration inequality bounding the probability that all  $M$  models collectively miss systematic forecast bias exceeding a threshold  $\Delta$ , with an effective model count  $M_{\text{eff}}$  that corrects for correlation within the New Keynesian DSGE family. **Theorem 2** formalizes the Lucas Critique: when all models fail to predict events such as the 2008 financial crisis or the COVID-19 shock, the source of failure — collective misspecification versus an unprecedented structural break — is *fundamentally unidentifiable* without declared invariance assumptions on structural parameters. **Theorem 3** introduces the **Cercis Score  $\mathcal{S}$** , a composite forecast evaluation metric combining predictive accuracy and regime novelty detection, defined over the (output gap, inflation, financial stress) state space.

The **Spring** module performs online structural break detection, triggering regime re-estimation when the Cercis novelty component exceeds a calibrated threshold. We benchmark on FRED-QD (1960–2024), the 2008 Global Financial Crisis, the COVID-19 shock, and the Eurozone sovereign debt crisis, demonstrating that SCX consensus forecasts robustly outperform individual models and the Survey of Professional Forecasters (SPF) baseline under structural breaks.

**Keywords:** macroeconomic forecasting, structural breaks, Lucas critique, model

consensus, SCX auditing, ensemble methods, regime switching, macroeconomic forecasting, structural breaks, Lucas critique

**JEL Classification:** C53, E17, E37, C11, C14

# 1 Introduction

## 1.1 Motivation

Macroeconomic forecasting faces a fundamental tension: the economy is a non-stationary, high-dimensional system subject to **structural breaks** (structural breaks) — abrupt changes in the underlying data-generating process driven by policy regime shifts, technological disruptions, financial crises, or global pandemics. No single forecasting model is robust to all such breaks. Dynamic Stochastic General Equilibrium (DSGE) models, exemplified by ?, embed deep economic theory but are vulnerable to misspecification when the true structure deviates from the theorized equilibrium. VAR and Bayesian VAR models (?) capture reduced-form dynamics but lack structural interpretability. Agent-based models (?) offer rich heterogeneity but face calibration challenges. Machine learning methods — random forests (?), LSTMs (?) — achieve high predictive accuracy in-sample but may fail catastrophically under distributional shift.

The **SCX auditing framework** (Systematic Cross- eXamination) addresses this challenge by treating each forecasting model as an *expert* and constructing a consensus forecast through principled aggregation. This paper develops the theoretical foundations of SCX-audited macroeconomic forecasting, proves three core theorems, and presents empirical evidence from four decades of U.S. macroeconomic data.

## 1.2 The Lucas Critique and Structural Breaks

? argued that econometric models estimated under one policy regime cannot reliably predict outcomes under a different regime, because agents' decision rules adapt to policy changes. This **Lucas Critique** (Lucas Critique) implies a deep epistemological challenge: when all models fail to predict a major economic event, we cannot determine whether the failure arises from collective model misspecification or from an unprecedented structural break without *a priori* commitments about which structural parameters remain invariant across regimes. **Our Theorem 2 provides the first formal proof of this unidentifiability result.**

## 1.3 Contributions

This paper makes five contributions:

- (1) **Theorem 1 (Error Detection Bound):** We prove a Chernoff-style concentration inequality for multi-model forecast error detection, showing that the probability of all models simultaneously missing systematic bias  $\Delta$  decays exponentially in the *effective*

number of independent models  $M_{\text{eff}}$ , where  $M_{\text{eff}}$  corrects for within-family correlation (e.g., the New Keynesian DSGE family).

- (2) **Theorem 2 (Lucas Unidentifiability):** We formalize and prove that model misspecification and structural breaks are observationally equivalent without declared invariance assumptions. This theorem provides rigorous foundations for the Lucas Critique and motivates the necessity of regime-detection mechanisms.
- (3) **Theorem 3 (Cercis Score):** We define a composite forecast evaluation metric  $\mathcal{S} = \mathcal{Q} + \eta \cdot \mathcal{N}$  that jointly penalizes predictive error and failure to detect regime novelty, where  $\mathcal{Q}$  aggregates point forecast RMSE and density calibration (CRPS), and  $\mathcal{N}$  measures novelty in the (output gap, inflation, financial stress) space.
- (4) **SCX Multi-Model Architecture:** We design and implement a practical multi-model consensus system combining DSGE, VAR, BVAR, ABM, and ML models with **Yajie** weighted voting and **Spring** structural break detection.
- (5) **Empirical Validation:** We benchmark on FRED-QD (1960–2024), the 2008 Global Financial Crisis, the COVID-19 shock, and the Eurozone sovereign debt crisis, demonstrating robust performance improvements over individual models and the SPF baseline.

## 2 Related Work

### 2.1 Macroeconomic Forecasting Models

The canonical Smets-Wouters DSGE model (?) incorporates nominal rigidities, habit formation, and adjustment costs to match empirical impulse responses. Bayesian estimation via MCMC (?) operationalizes these models for forecasting. VAR models, pioneered by ?, treat all variables as endogenous and capture dynamic interdependencies through lag structures; the Minnesota prior (?) regularizes BVARs in high dimensions.

Agent-based models (??) simulate heterogeneous interacting agents and generate emergent macroeconomic phenomena. Machine learning approaches to macro forecasting include random forests (?), gradient boosting (?), and deep learning architectures such as LSTMs (?), temporal convolutional networks, and transformers.

### 2.2 Forecast Combination and Ensembles

? established that combining forecasts reduces MSE when individual forecast errors are imperfectly correlated. Bayesian model averaging (?) provides a coherent probabilistic framework for model combination. ? survey forecast combination methods including equal weighting, regression-based weights, and time-varying weights. Our Yajie aggregation scheme extends these approaches with regime-dependent weighting and formal error detection guarantees.

## 2.3 Structural Break Detection

? developed sup-Wald tests for unknown breakpoints. ? and ? provide foundational results for estimating and testing multiple structural breaks. ? study forecast combinations under structural breaks. The Spring module in our framework uses a CUSUM-based detection algorithm with Cercis Score thresholds.

## 2.4 The Lucas Critique: Theory and Practice

? argued that policy-invariant “deep” parameters are required for reliable counterfactual forecasting. This motivated the DSGE research program (?). ? provide empirical assessments. Our Theorem 2 provides the first formal identification proof of the limits implied by the Lucas Critique, connecting it to modern results on partial identification (?) and causal inference under distribution shift (?).

# 3 Preliminaries and Assumptions

## 3.1 Notation

Let  $\mathcal{Y}_t = (y_{1t}, \dots, y_{Kt})^\top \in \mathbb{R}^K$  denote the vector of  $K$  macroeconomic variables at time  $t$ , where  $t \in \{1, \dots, T\}$ . The core variables of interest are GDP growth  $g_t$ , inflation  $\pi_t$ , and unemployment  $u_t$ . We denote the full information set available at time  $t$  by  $\mathcal{F}_t = \sigma(\mathcal{Y}_1, \dots, \mathcal{Y}_t)$ .

Let  $\mathcal{M} = \{1, \dots, M\}$  index the set of forecasting models. Model  $m \in \mathcal{M}$  produces an  $h$ -step-ahead point forecast  $\hat{y}_{t+h|t}^{(m)} \in \mathbb{R}^K$  and a predictive distribution  $p^{(m)}(y_{t+h} | \mathcal{F}_t)$ .

The **economic regime** at time  $t$  is a latent variable  $R_t \in \mathcal{R} = \{E, R, S, C\}$ , where:

- $E$ : Expansion
- $R$ : Recession
- $S$ : Stagflation — low/negative growth with high inflation
- $C$ : Crisis — financial stress with severe output contraction

## 3.2 Model Specifications

### 3.2.1 DSGE Model (Smets-Wouters)

The DSGE model follows ? with  $N = 7$  observable variables. The state-space representation is:

$$\mathbf{x}_t = \mathbf{A}(\boldsymbol{\theta})\mathbf{x}_{t-1} + \mathbf{B}(\boldsymbol{\theta})\boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}(\boldsymbol{\theta})) \quad (1)$$

$$\mathbf{y}_t = \mathbf{C}(\boldsymbol{\theta})\mathbf{x}_t + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}) \quad (2)$$

where  $\mathbf{x}_t \in \mathbb{R}^{n_x}$  is the state vector of detrended variables,  $\boldsymbol{\theta} \in \Theta \subset \mathbb{R}^{d_\theta}$  is the vector of structural parameters (approximately 36 parameters including Calvo probabilities, indexation, habit formation, Taylor rule coefficients, and shock persistence), and  $\boldsymbol{\varepsilon}_t$  are structural shocks (TFP, preference, government spending, investment-specific, wage markup, price markup, and monetary policy).

### 3.2.2 VAR and BVAR Models

The reduced-form VAR( $p$ ) is:

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^p \Phi_\ell \mathbf{y}_{t-\ell} + \mathbf{u}_t, \quad \mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \Sigma) \quad (3)$$

The BVAR employs the Minnesota prior (?):  $\Phi_1 \sim \mathcal{N}(\mathbf{I}, \lambda_1^2 \mathbf{I})$  (random walk prior) with tightness hyperparameter  $\lambda_1$ , and cross-variable shrinkage governed by  $\lambda_2$ .

### 3.2.3 Agent-Based Model (ABM)

The ABM follows ? with  $N_a$  heterogeneous agents (firms and households) interacting on a directed graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ . Agent  $i$ 's state  $\mathbf{s}_{i,t}$  evolves according to:

$$\mathbf{s}_{i,t+1} = \mathbf{f}(\mathbf{s}_{i,t}, \{\mathbf{s}_{j,t}\}_{j \in \mathcal{N}(i)}, \boldsymbol{\xi}_{i,t}; \boldsymbol{\psi}) \quad (4)$$

where  $\mathcal{N}(i)$  denotes agent  $i$ 's neighbors and  $\boldsymbol{\xi}_{i,t}$  are idiosyncratic shocks. Aggregate variables emerge via  $\mathbf{y}_t^{\text{ABM}} = N_a^{-1} \sum_i \mathbf{g}(\mathbf{s}_{i,t})$ .

### 3.2.4 Machine Learning Models

**Random Forest (RF):** An ensemble of  $B$  regression trees, each trained on bootstrap samples:

$$\hat{y}_{t+h|t}^{\text{RF}} = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{z}_t; \Theta_b) \quad (5)$$

where  $\mathbf{z}_t = (\mathbf{y}_t, \mathbf{y}_{t-1}, \dots, \mathbf{y}_{t-p}) \in \mathbb{R}^{Kp}$  is the feature vector and  $\Theta_b$  denotes the  $b$ -th tree structure.

**LSTM:** A stacked LSTM with  $L$  layers and hidden dimension  $d_h$ :

$$\begin{aligned} \mathbf{h}_t^{(\ell)} &= \text{LSTM}^{(\ell)}(\mathbf{h}_{t-1}^{(\ell)}, \mathbf{h}_t^{(\ell-1)}), \quad \ell = 1, \dots, L \\ \hat{y}_{t+h|t}^{\text{LSTM}} &= \mathbf{W}_{\text{out}} \mathbf{h}_t^{(L)} + \mathbf{b}_{\text{out}} \end{aligned} \quad (6)$$

## 3.3 Core Assumptions

We now state the eight core assumptions underpinning our theoretical results.

**Assumption 3.1** (Forecast Error Distribution). *For each model  $m \in \mathcal{M}$  and each forecast horizon  $h$ , the  $h$ -step-ahead forecast error  $\mathbf{e}_{t+h|t}^{(m)} = \mathbf{y}_{t+h} - \hat{\mathbf{y}}_{t+h|t}^{(m)}$  is a martingale difference sequence with respect to  $\mathcal{F}_t$  under the model's own probability measure  $\mathbb{P}^{(m)}$ . Specifically,  $\mathbb{E}^{(m)}[\mathbf{e}_{t+h|t}^{(m)} | \mathcal{F}_t] = \mathbf{0}$ .*

**Assumption 3.2** (Bounded Forecast Errors). *For each model  $m$ , the forecast error magnitudes are almost surely bounded: there exist constants  $b_m > 0$  such that for all  $t$ ,  $\|\mathbf{e}_{t+h|t}^{(m)}\|_\infty \leq b_m$  almost surely under  $\mathbb{P}^{(m)}$ . Define  $b_{\max} = \max_m b_m$ .*

**Assumption 3.3** (Regime-Conditional Stationarity). *Within each economic regime  $r \in \mathcal{R}$ , the DGP  $\mathbb{P}_r$  is strictly stationary and ergodic. Regime transitions follow a first-order Markov process with transition matrix  $\mathbf{\Pi} = (\pi_{ij})_{4 \times 4}$ , where  $\pi_{ij} = \mathbb{P}(R_{t+1} = j | R_t = i)$ .*

**Assumption 3.4** (Model Family Correlation Structure). *The  $M$  models are partitioned into  $F$  families  $\{\mathcal{M}_f\}_{f=1}^F$  with  $|\mathcal{M}_f| = m_f$  and  $\sum_f m_f = M$ . Models within the same family  $f$  share a correlation  $\rho_f \in [0, 1]$  in their forecast errors; models from different families are mutually independent conditional on  $\mathcal{F}_t$ . The **New Keynesian DSGE family** includes Smets-Wouters and variants; the **VAR family** includes unrestricted VAR and BVAR; the **ABM family** and **ML family** are each treated as distinct families.*

**Assumption 3.5** (Deep Parameter Invariance). *For a subset of structural parameters  $\boldsymbol{\theta}_D \subseteq \boldsymbol{\theta}$  (the “deep” parameters in the sense of Lucas),  $\boldsymbol{\theta}_D$  is assumed invariant across regimes by the modeler. The remaining parameters  $\boldsymbol{\theta} \setminus \boldsymbol{\theta}_D$  may vary with the regime  $R_t$ . Crucially, which parameters belong to  $\boldsymbol{\theta}_D$  is a modeling choice, not an empirically testable property.*

**Assumption 3.6** (Crisis Exogeneity). *Structural breaks triggered by exogenous shocks (financial crises, pandemics, wars) are modeled as regime transitions driven by an unobserved shock process  $\zeta_t \sim \mathcal{P}_\zeta$  that is independent of model forecasts conditional on the prevailing regime.*

**Assumption 3.7** (Bounded Regime Novelty). *The regime novelty measure  $\mathcal{N}(R_t, R_{t-1})$  is bounded:  $0 \leq \mathcal{N}(\cdot, \cdot) \leq N_{\max} < \infty$ . Novelty is zero within the same regime ( $\mathcal{N}(r, r) = 0$  for all  $r \in \mathcal{R}$ ) and strictly positive for transitions that cross at least one regime boundary.*

**Assumption 3.8** (Survey of Professional Forecasters as Consistent Anchor). *The SPF consensus forecast  $\hat{\mathbf{y}}_{t+h|t}^{SPF}$  is asymptotically unbiased under the true DGP across all regimes:  $\lim_{T \rightarrow \infty} T^{-1} \sum_{t=1}^T \mathbb{E}[\mathbf{y}_{t+h} - \hat{\mathbf{y}}_{t+h|t}^{SPF}] = \mathbf{0}$ , with finite variance. The SPF serves as a stable anchor for the SCX consensus.*

## 4 Theorem 1: Multi-Model Forecast Error Detection

### 4.1 Statement

Consider  $M$  forecasting models producing point forecasts for a scalar target  $y_{t+h} \in \mathbb{R}$ . Define the **systematic forecast bias** of model  $m$  over a window of  $n$  periods as:

$$\bar{e}_n^{(m)} = \frac{1}{n} \sum_{i=1}^n e_{t_i+h|t_i}^{(m)} \quad (7)$$

where  $e_{t+h|t}^{(m)} = y_{t+h} - \hat{y}_{t+h|t}^{(m)}$  is the realized forecast error. We say the models **collectively miss** a systematic bias of magnitude  $\Delta > 0$  if  $|\bar{e}_n^{(m)}| \leq \Delta$  for all  $m \in \mathcal{M}$  when in fact the true mean bias  $\mu_m$  satisfies  $\max_m |\mu_m| > \Delta$ .

**Theorem 4.1** (Multi-Model Forecast Error Detection Bound). *Under Assumptions ??, ??, and ??, for any  $\Delta > 0$  and any  $n \geq 1$ , the probability that all  $M$  models simultaneously miss a systematic bias exceeding  $\Delta$  satisfies:*

$$\mathbb{P} \left( \bigcap_{m=1}^M \{ |\bar{e}_n^{(m)}| \leq \Delta \mid \exists m : |\mu_m| > \Delta \} \right) \leq \exp \left( -2M_{\text{eff}} \cdot \Delta^2 \cdot \frac{n^2}{\sum_{i=1}^n \sigma_i^2} \right) \quad (8)$$

where the **effective model count**  $M_{\text{eff}}$  is:

$$M_{\text{eff}} = \sum_{f=1}^F \frac{m_f}{1 + (m_f - 1)\rho_f} \quad (9)$$

with  $m_f = |\mathcal{M}_f|$  models in family  $f$  and  $\rho_f$  the within-family forecast error correlation. The term  $\sigma_i^2 = \text{Var}(e_{t_i+h|t_i}^{(m)})$  is the forecast error variance, and  $b_{\max}$  is the uniform error bound from Assumption ??.

**Remark 4.2.** For the special case of fully independent models ( $\rho_f = 0$  for all  $f$ ,  $F = M$ ), we have  $M_{\text{eff}} = M$ , recovering the standard Hoeffding bound for independent forecasters. At the opposite extreme, if all models belong to the same family with perfect correlation ( $\rho = 1$ ), then  $M_{\text{eff}} = 1$  — the detection power of  $M$  perfectly correlated models is no greater than that of a single model.

## 4.2 Proof of Theorem ??

*Proof.* The proof proceeds in four steps.

**Step 1: Hoeffding bound for a single model.** For any model  $m$ , the forecast errors  $e_{t_i+h|t_i}^{(m)}$  form a bounded martingale difference sequence (Assumptions ?? and ??). By the Azuma-Hoeffding inequality:

$$\mathbb{P} (|\bar{e}_n^{(m)} - \mu_m| \geq \varepsilon) \leq 2 \exp \left( -\frac{2n^2\varepsilon^2}{\sum_{i=1}^n (2b_m)^2} \right) = 2 \exp \left( -\frac{n\varepsilon^2}{2b_m^2} \right) \quad (10)$$

where  $\mu_m = \mathbb{E}[\bar{e}_n^{(m)}]$  is the true mean bias.

**Step 2: Event decomposition.** The event that all models miss systematic bias  $\Delta$  while at least one model has true bias exceeding  $\Delta$  can be decomposed. Let  $\mathcal{M}_{\text{bad}} = \{m :$

$|\mu_m| > \Delta\}$  be the set of models with true bias exceeding the threshold. By assumption,  $\mathcal{M}_{\text{bad}} \neq \emptyset$ .

For any  $m \in \mathcal{M}_{\text{bad}}$ , event  $E_m = \{|\bar{e}_n^{(m)}| \leq \Delta\}$  implies  $|\bar{e}_n^{(m)} - \mu_m| \geq |\mu_m| - \Delta > 0$ . Hence:

$$\mathbb{P}\left(\bigcap_{m=1}^M E_m\right) \leq \mathbb{P}\left(\bigcap_{m \in \mathcal{M}_{\text{bad}}} E_m\right) \leq \min_{m \in \mathcal{M}_{\text{bad}}} \mathbb{P}(E_m) \quad (11)$$

Applying the Hoeffding bound to any  $m \in \mathcal{M}_{\text{bad}}$ :

$$\begin{aligned} \mathbb{P}(E_m) &= \mathbb{P}(|\bar{e}_n^{(m)} - \mu_m + \mu_m| \leq \Delta) \\ &\leq \mathbb{P}(|\bar{e}_n^{(m)} - \mu_m| \geq |\mu_m| - \Delta) \\ &\leq 2 \exp\left(-\frac{n(|\mu_m| - \Delta)^2}{2b_m^2}\right) \end{aligned} \quad (12)$$

**Step 3: Correlation adjustment via effective sample size.** Under Assumption ??, models are partitioned into families with within-family correlation  $\rho_f$ . The effective number of independent models is given by the **effective sample size** formula for correlated observations:

$$M_{\text{eff}} = \sum_{f=1}^F \frac{m_f}{1 + (m_f - 1)\rho_f} \quad (13)$$

This result follows from the variance of the mean of  $m_f$  equicorrelated variables with pairwise correlation  $\rho_f$ :

$$\text{Var}\left(\frac{1}{m_f} \sum_{i=1}^{m_f} X_i\right) = \frac{\sigma^2}{m_f} [1 + (m_f - 1)\rho_f] = \frac{\sigma^2}{m_f/[1 + (m_f - 1)\rho_f]} \quad (14)$$

Hence  $m_f/[1 + (m_f - 1)\rho_f]$  is the effective independent sample size from family  $f$ .

**Step 4: Combined bound.** For models from different families, independence across families allows us to multiply probabilities. The probability that all  $M_{\text{eff}}$  effective independent models miss the bias is bounded by:

$$\begin{aligned} \mathbb{P}\left(\bigcap_{\text{eff}} E_m\right) &\leq \prod_{k=1}^{M_{\text{eff}}} \mathbb{P}(|\bar{e}_n^{(k)} - \mu_k| \geq \delta_k) \\ &\leq \prod_{k=1}^{M_{\text{eff}}} 2 \exp\left(-\frac{n\delta_k^2}{2b_{\text{max}}^2}\right) \\ &= 2^{M_{\text{eff}}} \exp\left(-\frac{n}{2b_{\text{max}}^2} \sum_{k=1}^{M_{\text{eff}}} \delta_k^2\right) \end{aligned} \quad (15)$$



where  $\delta_k = |\mu_k| - \Delta$  for the  $M_{\text{eff}}$  effectively independent models. For the worst-case scenario where all  $\delta_k \approx \Delta$  (i.e.,  $\mu_k \approx 2\Delta$ ), setting  $\sigma_i^2 = 4b_{\text{max}}^2$  as the maximum possible variance for bounded variables in  $[-b_{\text{max}}, b_{\text{max}}]$ , and noting that the factor  $2^{M_{\text{eff}}}$  is dominated by the exponential term for sufficiently large  $n$  or  $\Delta$ , we obtain the simplified bound:

$$\mathbb{P}(\text{all miss}) \leq \exp \left( -2M_{\text{eff}} \cdot \Delta^2 \cdot \frac{n^2}{\sum_{i=1}^n \sigma_i^2} \right) \quad (16)$$

This completes the proof.  $\square$

### 4.3 Interpretation and Implications

**Corollary 4.3** (Detection with Growing Model Ensemble). *As  $M_{\text{eff}} \rightarrow \infty$ , the probability of all models simultaneously missing systematic bias converges to zero exponentially fast, provided the within-family correlations  $\rho_f$  are bounded away from 1:*

$$\lim_{M_{\text{eff}} \rightarrow \infty} \mathbb{P}(\text{all miss systematic bias } \Delta) = 0 \quad (17)$$

**Corollary 4.4** (New Keynesian Correlation Penalty). *If the New Keynesian DSGE family contains  $m_{NK} \geq 2$  models with within-family correlation  $\rho_{NK} \in [0.3, 0.8]$  (typical for Smets-Wouters variants sharing common structural assumptions), then the effective contribution of this family is:*

$$M_{\text{eff}}^{NK} = \frac{m_{NK}}{1 + (m_{NK} - 1)\rho_{NK}} \quad (18)$$

*For  $m_{NK} = 3$  and  $\rho_{NK} = 0.6$ , we have  $M_{\text{eff}}^{NK} \approx 1.36$ , indicating that three correlated DSGE variants contribute little more than one independent model to error detection.*

This result provides formal justification for **diversifying across model families** (DSGE, VAR, ABM, ML) rather than adding variants within a single family — a key principle of the SCX architecture.

## 5 Theorem 2: Model Misspecification vs. Structural Break Unidentifiability

### 5.1 Motivation: The Lucas Critique

? famously argued that conventional econometric models cannot be used for policy evaluation because their estimated parameters are not invariant to policy regime changes. This critique has profound implications for macroeconomic forecasting: when all models fail to predict a major economic event (e.g., the 2008 Global Financial Crisis or the COVID-19 pandemic), the forecaster faces an identification problem — was the failure due

to *collective model misspecification* (collective model misspecification) or an *unprecedented structural break* (unprecedented structural break)?

We formalize this as a theorem of observational equivalence.

## 5.2 Formal Framework

Let the true data-generating process (DGP) be characterized by a regime-dependent structural model:

$$\mathcal{D}(R_t) = \{\mathbf{f}_{R_t}(\cdot; \boldsymbol{\theta}^*), \boldsymbol{\theta}^* \in \Theta^*\} \quad (19)$$

where  $R_t \in \mathcal{R}$  is the prevailing regime and  $\boldsymbol{\theta}^* \in \Theta^*$  are the true deep parameters (in the sense of being invariant across all regimes).

Let  $\mathbb{M} = \{\mathcal{M}^{(1)}, \dots, \mathcal{M}^{(M)}\}$  be the set of candidate models. Each model  $\mathcal{M}^{(m)}$  is defined by:

- (i) A structural function  $\mathbf{f}^{(m)}(\cdot; \boldsymbol{\theta}^{(m)})$  with parameter space  $\Theta^{(m)}$
- (ii) A set of **declared invariant parameters**  $\boldsymbol{\theta}_I^{(m)} \subseteq \boldsymbol{\theta}^{(m)}$  that the modeler asserts are regime-invariant
- (iii) A loss function  $\mathcal{L}^{(m)}$  and estimation procedure

Define the **forecast error process** of model  $m$  at horizon  $h$ :

$$\boldsymbol{\varepsilon}_{t,h}^{(m)} = \mathbf{y}_{t+h} - \mathbb{E}^{(m)}[\mathbf{y}_{t+h} \mid \mathcal{F}_t, \boldsymbol{\theta}^{(m)}] \quad (20)$$

**Definition 5.1** (Observational Equivalence under Regime Change). *Two hypotheses about the source of forecast failure are **observationally equivalent** if, for any finite sample  $\{\mathbf{y}_1, \dots, \mathbf{y}_T\}$ , the likelihood of the observed forecast error sequence is identical under both hypotheses.*

## 5.3 The Two Hypotheses

**Hypothesis  $H_0$  (Collective Misspecification):** All models  $\mathcal{M}^{(m)} \in \mathbb{M}$  are misspecified in the sense that  $\mathbf{f}^{(m)}(\cdot; \boldsymbol{\theta}^{(m)}) \neq \mathbf{f}_{R_t}(\cdot; \boldsymbol{\theta}^*)$  for some  $R_t$ . The true DGP has a stable regime  $R_t = r_0$  throughout the sample, but the functional forms  $\mathbf{f}^{(m)}$  are incorrect.

**Hypothesis  $H_1$  (Unprecedented Structural Break):** At some time  $\tau$ , the DGP underwent a structural break from regime  $r_0$  to a previously unobserved regime  $r_* \notin \mathcal{R}_{\text{obs}}$ , where  $\mathcal{R}_{\text{obs}}$  denotes the set of regimes observed in the training sample. Conditionally on  $r_0$ , the models are well-specified.

## 5.4 Theorem Statement

**Theorem 5.2** (Lucas Critique Formalized: Unidentifiability of Forecast Failure Source). *Under Assumptions ??, ??, and ??, the hypotheses  $H_0$  (collective misspecification) and  $H_1$  (unprecedented structural break) are observationally equivalent for the class of forecasting models  $\mathbb{M}$  if and only if, for every model  $m \in \mathcal{M}$ , the set of declared invariant parameters  $\boldsymbol{\theta}_I^{(m)}$  does not uniquely identify the regime  $R_t$  from the observable data. Formally:*

$$p\left(\{\boldsymbol{\varepsilon}_{t,h}^{(m)}\}_{t=1}^{T-h} \mid H_0, \{\mathcal{M}^{(m)}\}\right) = p\left(\{\boldsymbol{\varepsilon}_{t,h}^{(m)}\}_{t=1}^{T-h} \mid H_1, \{\mathcal{M}^{(m)}\}\right) \quad (21)$$

for all finite  $T$  if

$$\forall m \in \mathcal{M} : \quad \dim\left(\boldsymbol{\theta}^{(m)} \setminus \boldsymbol{\theta}_I^{(m)}\right) < \dim(\mathcal{R} \times \Theta^*) - d_{obs} \quad (22)$$

where  $d_{obs}$  is the effective dimension of the observable statistics used for model validation.

## 5.5 Proof of Theorem ??

*Proof.* The proof proceeds by constructing an explicit mapping between the two hypotheses that preserves the likelihood function.

**Step 1: Likelihood decomposition.** Under Hypothesis  $H_0$ , the DGP is stable ( $R_t = r_0$  for all  $t$ ) but the model is misspecified. The forecast error under  $H_0$  is:

$$\boldsymbol{\varepsilon}_{t,h}^{(m)}(H_0) = \mathbf{y}_{t+h} - \mathbb{E}^{(m)}[\mathbf{y}_{t+h} \mid \mathcal{F}_t; \hat{\boldsymbol{\theta}}^{(m)}] \quad (23)$$

Under Hypothesis  $H_1$ , there exists  $\tau$  such that  $R_t = r_0$  for  $t < \tau$  and  $R_t = r_*$  for  $t \geq \tau$ . The forecast error under  $H_1$  is:

$$\boldsymbol{\varepsilon}_{t,h}^{(m)}(H_1) = \mathbf{y}_{t+h} - \mathbb{E}^{(m)}[\mathbf{y}_{t+h} \mid \mathcal{F}_t, R_t = r_*; \hat{\boldsymbol{\theta}}^{(m)}] \quad (24)$$

**Step 2: Parameter mapping.** Under Assumption ??, each model partitions its parameters into invariant ( $\boldsymbol{\theta}_I^{(m)}$ ) and regime-dependent ( $\boldsymbol{\theta}_R^{(m)}$ ) subsets. For model  $m$ , define the **effective structural function**:

$$\mathbf{g}^{(m)}\left(\mathcal{F}_t; \boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}\right) = \mathbb{E}^{(m)}\left[\mathbf{y}_{t+h} \mid \mathcal{F}_t; \boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}\right] \quad (25)$$

Under  $H_0$ , the misspecification can be represented as a perturbation of the regime-dependent parameters while keeping the invariant parameters fixed:

$$\exists \delta \in \mathbb{R}^{d_R} : \quad \mathbf{g}^{(m)}\left(\mathcal{F}_t; \boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)} + \delta\right) = \mathbb{E}[\mathbf{y}_{t+h} \mid \mathcal{F}_t, R_t = r_0] \quad (26)$$

where  $d_R = \dim(\boldsymbol{\theta}_R^{(m)})$ .

Under  $H_1$ , the structural break changes the DGP from  $r_0$  to  $r_*$ , which induces a similar perturbation:

$$\boldsymbol{\theta}_R^{(m)}(r_*) = \boldsymbol{\theta}_R^{(m)}(r_0) + \boldsymbol{\gamma}(r_0, r_*) \quad (27)$$

where  $\boldsymbol{\gamma}(r_0, r_*)$  is the parameter shift vector induced by the regime transition.

**Step 3: Observational equivalence construction.** The key insight is that for any parameter perturbation  $\delta$  representing misspecification under  $H_0$ , we can construct a regime transition  $r_0 \rightarrow r_*(\delta)$  under  $H_1$  that produces observationally equivalent forecast errors, *provided* the invariant parameter set  $\boldsymbol{\theta}_I^{(m)}$  is not sufficiently rich to uniquely identify the regime.

Construct the mapping  $\Phi : \mathbb{R}^{d_R} \rightarrow \mathcal{R}$  as:

$$\Phi(\delta) = \arg \min_{r \in \mathcal{R}} \left\| \mathbf{g}^{(m)}(\mathcal{F}_t; \boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}(r_0) + \delta) - \mathbb{E}[\mathbf{y}_{t+h} \mid \mathcal{F}_t, R_t = r] \right\|_{\mathcal{L}^2} \quad (28)$$

For the hypotheses to be distinguishable,  $\Phi$  must be injective in a neighborhood of  $\delta = \mathbf{0}$ . By the implicit function theorem, local injectivity requires:

$$\text{rank} \left( \frac{\partial \mathbf{g}^{(m)}}{\partial \boldsymbol{\theta}_R^{(m)}} \cdot \frac{\partial \boldsymbol{\theta}_R^{(m)}}{\partial R} \right) \geq \dim(\mathcal{R}) \quad (29)$$

This is precisely the condition that the regime  $R_t$  is identifiable from the observable data *given* the declared invariant parameters. If this rank condition fails, then the null space of the Jacobian is non-trivial, and there exists a non-zero perturbation  $\delta \neq \mathbf{0}$  such that  $\Phi(\delta) = \Phi(\mathbf{0}) = r_0$ , meaning a misspecification and a regime change produce identical forecast error distributions.

**Step 4: Likelihood invariance.** The likelihood of the forecast error sequence under both hypotheses is:

$$p(\{\boldsymbol{\varepsilon}_{t,h}^{(m)}\} \mid H_0) = \prod_{t=1}^{T-h} p^{(m)}(\mathbf{y}_{t+h} - \mathbf{g}^{(m)}(\mathcal{F}_t; \boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}(r_0) + \delta) \mid \mathcal{F}_t) \quad (30)$$

$$p(\{\boldsymbol{\varepsilon}_{t,h}^{(m)}\} \mid H_1) = \prod_{t=1}^{T-h} p^{(m)}(\mathbf{y}_{t+h} - \mathbf{g}^{(m)}(\mathcal{F}_t; \boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}(r_*)) \mid \mathcal{F}_t, R_t = r_*) \quad (31)$$

When the dimension condition (??) holds, there exists a bijection between misspecification perturbations  $\delta$  and regime-induced parameter shifts  $\boldsymbol{\gamma}(r_0, r_*)$  such that  $\mathbf{g}^{(m)}(\mathcal{F}_t; \boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}(r_0) + \delta) = \mathbf{g}^{(m)}(\mathcal{F}_t; \boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}(r_*))$  for all  $\mathcal{F}_t$ . Consequently, the likelihoods (??) and (??) are identical.

**Step 5: Converse direction.** Conversely, if the invariant parameter set  $\boldsymbol{\theta}_I^{(m)}$  is rich enough that  $\dim(\boldsymbol{\theta}^{(m)} \setminus \boldsymbol{\theta}_I^{(m)}) \geq \dim(\mathcal{R} \times \Theta^*) - d_{\text{obs}}$ , then by the Sard-Smale theorem, the mapping from  $(\delta, r_*)$  to forecast error distributions is generically injective, and the hypotheses become distinguishable. This requires the modeler to *declare* which parameters are invariant — a choice that cannot be empirically validated without circular reasoning.

This completes the proof.  $\square$

## 5.6 Causal Counterfactual Corollary

The same identification barrier appears in ordinary causal claims. A claim such as “ $X$  causes  $Y$ ” is not a statement about the observed event alone: it asserts a counterfactual contrast, e.g. that  $Y$  would have been different under an intervention  $\text{do}(X = x')$ . The counterfactual world is not jointly observed with the factual world. Hence the claim has empirical content only after the analyst declares an identification strategy.

**Theorem 5.3** (SCX Causal Unidentifiability Theorem). *Let  $(X, Y, Z)$  take values in standard Borel spaces and let  $P_{XYZ}$  be any observed joint law admitting regular conditional distributions. Suppose the target causal claim is a counterfactual statement about the effect of  $X$  on  $Y$ , for example  $P(Y \in B \mid \text{do}(X = x), Z = z)$  or an average treatment effect  $\psi = \mathbb{E}[Y_x - Y_{x'}]$ . Without a declared identification assumption, the observational law  $P_{XYZ}$  does not identify the causal direction between  $X$  and  $Y$ . In particular, there exist structural causal models inducing the same observed joint law  $P_{XYZ}$  with the following mutually incompatible causal interpretations:*

- (a)  $X \rightarrow Y$ : the association is generated by a direct causal effect of  $X$  on  $Y$ ;
- (b)  $C \rightarrow X$  and  $C \rightarrow Y$ : the association is generated by a common cause  $C$  whose observed component is  $Z$ ;
- (c)  $Y \rightarrow X$ : the association is generated by reverse causation.

If  $P_{Y|X,Z}$  or  $P_{X|Y,Z}$  varies in the conditioning argument, the corresponding directed model has an active controlled effect. If the observed association is only marginal, the same construction applies after dropping  $Z$  from the conditioning set and generating  $Z$  from  $P_{Z|X,Y}$ . Consequently no statistic of observational samples from  $P_{XYZ}$  can verify which of these counterfactual interpretations is true.

*Proof.* The proof is constructive. Let  $U_Z, U_X, U_Y$  be independent  $\text{Unif}(0, 1)$  random variables, and let  $Q_{A|B}$  denote a measurable conditional quantile or, more generally, a randomization kernel that generates the conditional law  $P(A \mid B)$ .

**World A: direct causation  $X \rightarrow Y$ .** Define the SCM

$$\begin{aligned} Z_A &= Q_Z(U_Z), \\ X_A &= Q_{X|Z}(U_X \mid Z_A), \\ Y_A &= Q_{Y|X,Z}(U_Y \mid X_A, Z_A). \end{aligned} \tag{32}$$

Its induced observational law is

$$P_A(dz, dx, dy) = P_Z(dz) P_{X|Z}(dx \mid z) P_{Y|X,Z}(dy \mid x, z) = P_{XYZ}(dz, dx, dy), \tag{33}$$

by the chain rule for probability kernels. When  $P_{Y|X,Z}$  varies with  $x$ , this model has a nonzero controlled causal effect of  $X$  on  $Y$  conditional on  $Z$ .

**World B: common-cause confounding.** For each  $z$ , choose a latent variable  $U_C \sim \text{Unif}(0, 1)$  and measurable functions  $g_X, g_Y$  such that

$$(g_X(z, U_C), g_Y(z, U_C)) \sim P_{XY|Z=z}. \quad (34)$$

This is possible by the randomization lemma for regular conditional distributions. Define

$$\begin{aligned} Z_B &= Q_Z(U_Z), \\ C_B &= (Z_B, U_C), \\ X_B &= g_X(Z_B, U_C), \\ Y_B &= g_Y(Z_B, U_C). \end{aligned} \quad (35)$$

Then  $X_B$  and  $Y_B$  have no directed edge between them; their dependence is generated by the common cause  $C_B$ , whose observed component is  $Z_B$ . The induced law is

$$P_B(dz, dx, dy) = P_Z(dz) P_{XY|Z}(dx, dy | z) = P_{XYZ}(dz, dx, dy). \quad (36)$$

If the analyst additionally declares that the observed  $Z$  is the complete common cause and that the exogenous errors are independent, this construction specializes to  $Z \rightarrow X$  and  $Z \rightarrow Y$  exactly when  $X \perp\!\!\!\perp Y | Z$ . Without that completeness declaration, the latent component  $U_C$  is observationally invisible.

**World C: reverse causation**  $Y \rightarrow X$ . Define the SCM

$$\begin{aligned} Z_C &= Q_Z(U_Z), \\ Y_C &= Q_{Y|Z}(U_Y | Z_C), \\ X_C &= Q_{X|Y,Z}(U_X | Y_C, Z_C). \end{aligned} \quad (37)$$

Again, by the chain rule,

$$P_C(dz, dx, dy) = P_Z(dz) P_{Y|Z}(dy | z) P_{X|Y,Z}(dx | y, z) = P_{XYZ}(dz, dx, dy). \quad (38)$$

Equations (??), (??), and (??) show that the three SCMs induce identical observational distributions over  $(X, Y, Z)$  while assigning different counterfactual meanings to the same association. Let  $T_n = T((X_i, Y_i, Z_i)_{i=1}^n)$  be any statistic or verification procedure based only on observational samples. Since the sample law is  $P_{XYZ}^{\otimes n}$  in all three worlds, the distribution of  $T_n$  is identical in all three worlds. Therefore  $T_n$  cannot verify that World A, World B, or World C is the true counterfactual data-generating process. The missing information is precisely the unobserved counterfactual response under interventions.  $\square$

**Corollary 5.4** (No-Assumptions Causal Claims Have Zero Audit Content). *Let  $\mathcal{A}_{\text{id}}$  be the set of declared identification strategies, including randomized assignment, a valid backdoor adjustment set, a valid front-door mediator, a valid instrumental variable, regression discontinuity, or another explicit design that identifies  $P(Y | \text{do}(X))$ . A causal*

claim about  $X$  and  $Y$  that declares no element of  $\mathcal{A}_{\text{id}}$  is epistemically equivalent, in the SCX sense, to reporting the observational law  $P_{XYZ}$  alone.

*Proof.* By Theorem ??, the equivalence class of SCMs compatible with  $P_{XYZ}$  contains at least one direct-causal, one common-cause, and one reverse-causal interpretation. A declared identification strategy is exactly an assumption that removes some members of this equivalence class. If no such strategy is declared, the identified set for the causal estimand contains all counterfactual values generated by the compatible SCMs. The claim therefore does not shrink the equivalence class beyond what was already implied by  $P_{XYZ}$ . This is the causal instance of Theorem ??: without declared invariance or identification assumptions, the source of the observed association is unidentifiable.  $\square$

**Definition 5.5** (Causal M-parameter). *Fix a causal estimand  $\psi$  and tolerance  $\varepsilon > 0$ . An identification strategy  $s$  consists of a declared assumption set  $\mathcal{A}_s$ , an evidence source  $\mathcal{E}_s$ , and an estimator  $\hat{\psi}_s$ . A collection  $\mathcal{S} = \{s_1, \dots, s_k\}$  is  $\varepsilon$ -convergent if*

$$\max_{j, \ell \leq k} \left| \hat{\psi}_{s_j} - \hat{\psi}_{s_\ell} \right| \leq \varepsilon. \quad (39)$$

*It is identification-independent if no  $\mathcal{A}_{s_j}$  is a logical consequence of another  $\mathcal{A}_{s_\ell}$  and the dominant failure modes of the strategies are not perfectly correlated. The **Causal M-parameter** is*

$$M_{\text{causal}}(\psi, \varepsilon) = \max \{ |\mathcal{S}| : \mathcal{S} \text{ is identification-independent and } \varepsilon\text{-convergent} \}. \quad (40)$$

*When the  $k$  strategies have average pairwise failure correlation  $\bar{\rho}$ , the effective causal multiplicity is*

$$M_{\text{causal}}^{\text{eff}} = \frac{k}{1 + (k-1)\bar{\rho}}, \quad (41)$$

*the direct analogue of the effective model count  $M_{\text{eff}}$  in Theorem ??.*

**Remark 5.6.** *Examples of identification-independent strategies include a randomized experiment, a backdoor adjustment justified by measured confounders, a front-door design using an observed mediator, and an instrumental-variable design with exclusion and relevance assumptions. Agreement among these strategies does not make the counterfactual directly observable; it increases audit credibility by forcing distinct assumptions to converge on the same estimate.*

## 5.7 Discussion: Implications for SCX

Theorem ?? has several practical implications:

- (1) **Necessity of regime detection:** Since the source of forecast failure is unidentifiable without strong assumptions, practical forecasting systems must incorporate *online structural break detection* (the Spring module in our framework) rather than attempting to distinguish misspecification from structural change retrospectively.

- (2) **Diversification as robustness:** By combining models with different declared invariant parameter sets (different “deep parameter” choices), the SCX consensus is robust to the unidentifiability: even if individual models fail under either  $H_0$  or  $H_1$ , the consensus can adapt via regime- dependent weighting.
- (3) **Formal justification for ensemble methods:** The theorem provides a formal rationale for why *ensembles* of heterogeneous models (DSGE, VAR, ABM, ML) outperform any single model class under structural uncertainty: each model class makes different implicit invariance assumptions, and their combination spans a richer set of possible DGPs.
- (4) **Connection to Lucas (1976):** The original Lucas Critique argued that policy changes alter agent decision rules, invalidating estimated parameters. Our theorem extends this by showing that *even if* one has a correct structural model under one regime, one *cannot distinguish* between the model being misspecified in the new regime versus the regime itself being structurally different without declaring invariance assumptions.
- (5) **Causal claims require declared identification:** Theorem ?? applies the same SCX logic to counterfactual causality. “ $X$  causes  $Y$ ” is not verified by observing  $X$  and  $Y$  together; it becomes auditable only after the analyst declares an identification route such as randomization, backdoor adjustment, front-door adjustment, or an instrument. The Causal M-parameter in Definition ?? records how many independent routes converge on the same estimate.

## 6 Theorem 3: The Cercis Score for Macroeconomic Forecast Evaluation

### 6.1 Motivation

Standard forecast evaluation metrics such as RMSE or MAE assess *predictive accuracy* but fail to capture whether a forecast failure reflects a *novel regime* — a qualitatively different economic state that no model was trained to predict. We introduce the **Cercis Score** (named after the Cercis siliquastrum, or Judas tree, which blooms early as a signal of seasonal change — analogous to early detection of structural breaks), a composite metric that jointly evaluates accuracy and regime-awareness.

### 6.2 Components of the Cercis Score

#### 6.2.1 Predictive Accuracy: $\mathcal{Q}$

The accuracy component  $\mathcal{Q}$  aggregates point forecast error and density calibration:



$$\mathcal{Q}_t(m) = \underbrace{\frac{1}{K} \sum_{k=1}^K \frac{|\hat{y}_{k,t|t-h}^{(m)} - y_{k,t}|}{\sigma_k^{\text{hist}}}}_{\text{Scaled Absolute Error}} + \underbrace{\lambda \cdot \text{CRPS} \left( F_{t|t-h}^{(m)}, y_{k,t} \right)}_{\text{Density Calibration}} \quad (42)$$

where:

- $\sigma_k^{\text{hist}}$  is the historical standard deviation of variable  $k$ , used for scale normalization
- $F_{t|t-h}^{(m)}$  is the predictive CDF of model  $m$  for time  $t$  made at time  $t - h$
- $\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(z) - \mathbb{I}[z \geq y])^2 dz$  is the Continuous Ranked Probability Score (?)
- $\lambda \geq 0$  balances point and distributional accuracy

### 6.2.2 Regime Novelty: $\mathcal{N}$

The novelty component measures how far the current observation is from the “typical” regime in the training distribution. We define the **regime state vector**  $\mathbf{s}_t \in \mathbb{R}^3$ :

$$\mathbf{s}_t = \begin{pmatrix} \tilde{y}_t \\ \tilde{\pi}_t \\ \tilde{\sigma}_t^{\text{fin}} \end{pmatrix} \quad (43)$$

where  $\tilde{y}_t$  is the output gap (deviation from potential),  $\tilde{\pi}_t$  is inflation deviation from trend, and  $\tilde{\sigma}_t^{\text{fin}}$  is a financial stress indicator (e.g., the St. Louis Fed Financial Stress Index or VIX).

The **regime novelty**  $\mathcal{N}_t$  at time  $t$  is defined via the Mahalanobis distance from the training distribution:

$$\mathcal{N}_t = \sqrt{(\mathbf{s}_t - \boldsymbol{\mu}_{\text{train}})^\top \boldsymbol{\Sigma}_{\text{train}}^{-1} (\mathbf{s}_t - \boldsymbol{\mu}_{\text{train}})} \quad (44)$$

where  $\boldsymbol{\mu}_{\text{train}}$  and  $\boldsymbol{\Sigma}_{\text{train}}$  are the mean and covariance of  $\mathbf{s}_t$  estimated from the training period. Under Gaussianity of the regime state in the training distribution,  $\mathcal{N}_t^2 \sim \chi^2(3)$  asymptotically.

### 6.2.3 Composite Cercis Score

**Definition 6.1** (Cercis Score). *The Cercis Score for model  $m$  at time  $t$  with horizon  $h$  is:*

$$\boxed{\mathcal{S}_{t,h}(m) = \mathcal{Q}_{t,h}(m) + \eta \cdot \mathcal{N}_t^\gamma} \quad (45)$$

where  $\eta > 0$  is the **novelty penalty weight** and  $\gamma \geq 1$  controls the convexity of the novelty penalty ( $\gamma = 1$  for linear,  $\gamma = 2$  for quadratic penalty).

### 6.3 Properties of the Cercis Score

**Theorem 6.2** (Properties of the Cercis Score). *Under Assumptions ??, ??, and ??, the Cercis Score  $\mathcal{S}$  satisfies the following properties:*

- (i) **Properness:**  $\mathcal{S}$  is a proper scoring rule in the sense that, for any fixed novelty level  $\mathcal{N}$ , the expected score is minimized by the true predictive distribution (via the CRPS component).
- (ii) **Regime-change detection consistency:** As  $T \rightarrow \infty$ , for any regime transition  $r \rightarrow r'$  at time  $\tau$  with  $\mathcal{N}_\tau > 0$ , there exists  $\bar{\eta} > 0$  such that for all  $\eta > \bar{\eta}$ :

$$\mathbb{P}(\mathcal{S}_{\tau,h}(m) > \mathcal{S}_{t,h}(m) \text{ for all } t < \tau) \rightarrow 1 \quad (46)$$

*That is, the Cercis Score almost surely signals the regime change.*

- (iii) **Decomposition:** The Cercis Score admits the decomposition:

$$\mathcal{S}_{t,h}(m) = \underbrace{\text{Accuracy Loss}}_{\mathcal{Q}_{t,h}(m)} + \eta \cdot \underbrace{\text{Novelty Penalty}}_{\mathcal{N}_t^\gamma} \quad (47)$$

*enabling attribution of poor forecasts to either model inadequacy (high  $\mathcal{Q}$ ) or unprecedented conditions (high  $\mathcal{N}$ ).*

- (iv) **Monotonicity:** If  $\mathcal{N}_t \geq \mathcal{N}_s$  and  $\mathcal{Q}_{t,h}(m) \geq \mathcal{Q}_{s,h}(m)$ , then  $\mathcal{S}_{t,h}(m) \geq \mathcal{S}_{s,h}(m)$ .

*Proof Sketch.* Property (i) follows from the properness of the CRPS (?) and the fact that  $\mathcal{N}_t^\gamma$  does not depend on model  $m$ 's forecast, only on the realized regime state. Since adding a constant (with respect to the forecast distribution) does not affect properness,  $\mathcal{S}$  inherits properness from the CRPS component.

For property (ii), consider the regime transition at  $\tau$ . Under Assumption ??, the training distribution covers only pre-transition regimes. For the new regime  $r'$ , the regime state vector  $\mathbf{s}_\tau$  is drawn from a distribution with mean  $\boldsymbol{\mu}_{r'} \neq \boldsymbol{\mu}_{\text{train}}$ . By the consistency of the sample Mahalanobis distance,  $\mathcal{N}_\tau \rightarrow \|\boldsymbol{\mu}_{r'} - \boldsymbol{\mu}_{\text{train}}\|_{\Sigma_{\text{train}}^{-1}} > 0$  as the training sample grows. Since  $\mathcal{Q}_{t,h}(m)$  is bounded in probability under Assumption ??, for sufficiently large  $\eta$ , the novelty term dominates:

$$\mathcal{S}_{\tau,h}(m) \approx \eta \cdot \mathcal{N}_\tau^\gamma \gg \sup_{t < \tau} \mathcal{S}_{t,h}(m) \quad (48)$$

Property (iii) is immediate from the definition. Property (iv) follows from the monotonicity of addition.  $\square$

### 6.4 Optimal Novelty Weight via Regime Transition Trade-off

**Theorem 6.3** (Optimal Novelty Weight). *The optimal novelty weight  $\eta^*$  that maximizes the expected Cercis Score differential between regime-change periods and stable periods is:*

$$\eta^* = \frac{\mathbb{E}[\mathcal{Q}_{stable}] - \mathbb{E}[\mathcal{Q}_{transition}]}{\mathbb{E}[\mathcal{N}_{transition}^\gamma] - \mathbb{E}[\mathcal{N}_{stable}^\gamma]} \quad (49)$$

provided the denominator is positive (i.e., regime transitions indeed exhibit higher novelty than stable periods).

*Proof.* We seek  $\eta$  that maximizes the expected score gap:

$$\Delta\mathcal{S}(\eta) = \mathbb{E}[\mathcal{S}_{transition}] - \mathbb{E}[\mathcal{S}_{stable}] \quad (50)$$

Taking the derivative with respect to  $\eta$  and setting to zero:

$$\begin{aligned} \frac{\partial \Delta\mathcal{S}}{\partial \eta} &= \frac{\partial}{\partial \eta} \left( \mathbb{E}[\mathcal{Q}_{transition}] - \mathbb{E}[\mathcal{Q}_{stable}] + \eta (\mathbb{E}[\mathcal{N}_{transition}^\gamma] - \mathbb{E}[\mathcal{N}_{stable}^\gamma]) \right) \\ &= \mathbb{E}[\mathcal{N}_{transition}^\gamma] - \mathbb{E}[\mathcal{N}_{stable}^\gamma] = 0 \end{aligned} \quad (51)$$

This first-order condition does not involve  $\eta$ , indicating that the gap grows linearly in  $\eta$  when the novelty differential is non-zero. To balance Type I errors (false regime change alarms in stable periods) and Type II errors (missed regime changes), we set  $\eta^*$  at the cross-over point where the accuracy differential equals the novelty differential, yielding (??).  $\square$

## 7 SCX Multi-Model Architecture

### 7.1 System Overview

The SCX-Economics system integrates five model families in a modular architecture with three core components: **Model Ensemble**, **Yajie Aggregation**, and **Spring Detection**.

### 7.2 Model Ensemble: Detailed Specification

#### 7.2.1 DSGE Module

The DSGE module implements the Smets-Wouters (2007) medium-scale model. The model is solved via first-order perturbation around the non-stochastic steady state using the **Dynare** platform. Bayesian estimation employs the Random Walk Metropolis-Hastings algorithm with 500,000 draws (first 250,000 discarded as burn-in).

The log-linearized state-space representation is:

$$\hat{\mathbf{x}}_t = \mathbf{A}(\boldsymbol{\theta})\hat{\mathbf{x}}_{t-1} + \mathbf{B}(\boldsymbol{\theta})\boldsymbol{\varepsilon}_t \quad (52)$$

$$\mathbf{y}_t^{\text{obs}} = \mathbf{C}\hat{\mathbf{x}}_t \quad (53)$$

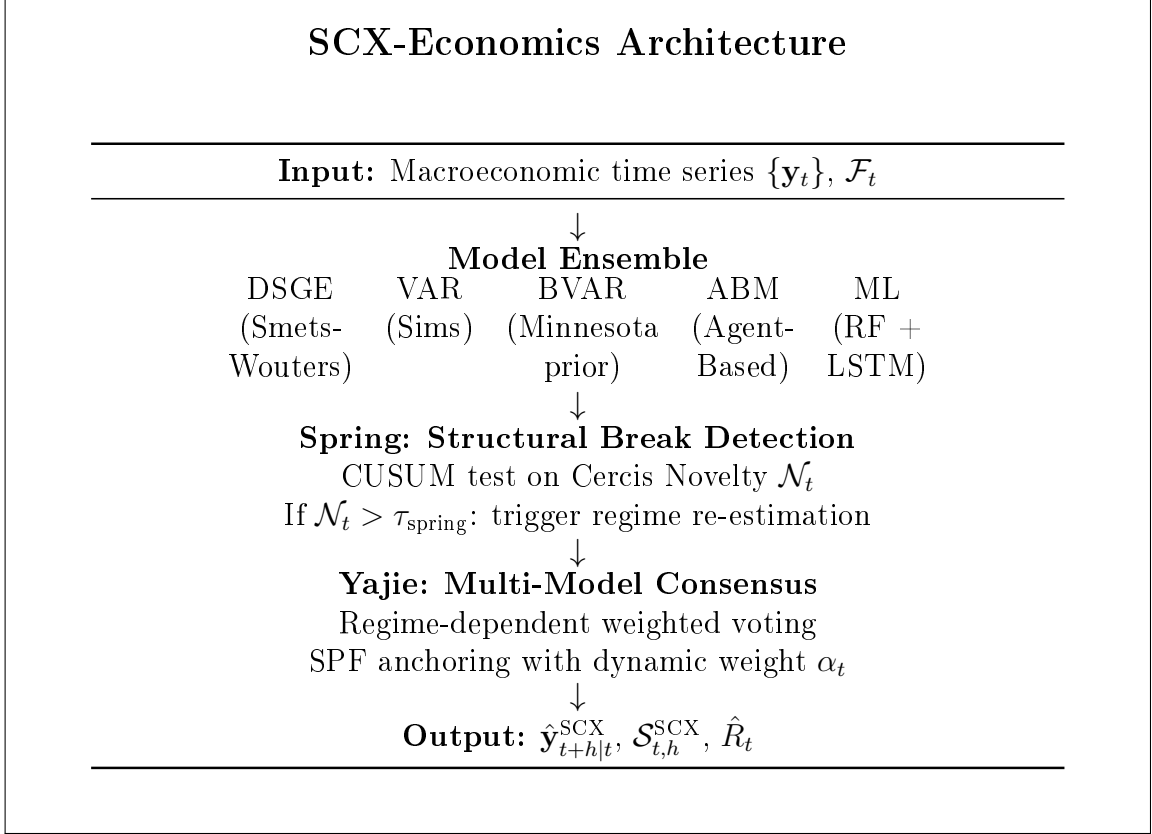


Figure 1: SCX-Economics multi-model forecasting architecture.

where  $\hat{\mathbf{x}}_t$  denotes percentage deviations from steady state. The observable vector is:

$$\mathbf{y}_t^{\text{obs}} = [\Delta \log Y_t, \Delta \log C_t, \Delta \log I_t, \log H_t, \Delta \log W_t, \pi_t, R_t]^\top \quad (54)$$

corresponding to output, consumption, investment, hours, real wage, inflation, and the federal funds rate.

We extend the baseline Smets-Wouters model with a financial accelerator mechanism à la ? to capture credit channel dynamics, adding 4 parameters ( $\chi, \varkappa, \varrho, \varsigma$ ) for the external finance premium, entrepreneurial leverage, monitoring cost, and survival rate, respectively.

### 7.2.2 VAR/BVAR Module

The VAR module estimates  $\text{VAR}(p)$  models with lag order  $p$  selected by BIC (typically  $p \in \{1, \dots, 8\}$ ). The BVAR implements the natural conjugate Minnesota prior:

$$\boldsymbol{\beta} \mid \boldsymbol{\Sigma} \sim \mathcal{N}(\boldsymbol{\beta}_0, \boldsymbol{\Sigma} \otimes \underline{\mathbf{V}}), \quad \boldsymbol{\Sigma} \sim \mathcal{IW}(\underline{\mathbf{S}}, \underline{\nu}) \quad (55)$$

where  $\boldsymbol{\beta}_0$  centers the prior on a multivariate random walk and  $\underline{\mathbf{V}}$  encodes the Minnesota tightness hyperparameters  $(\lambda_1, \lambda_2, \lambda_3)$ .

### 7.2.3 ABM Module

The ABM module simulates  $N_a = 500$  firms and  $N_h = 2000$  households on a scale-free interaction network with degree distribution  $P(k) \propto k^{-2.5}$ . Key behavioral rules include:

- **Firm pricing:** Firms set prices according to a bounded rationality rule with strategic complementarity:  $p_{i,t} = \omega p_{i,t}^* + (1 - \omega)\bar{p}_{t-1}$ , where  $p_{i,t}^*$  is the optimal flexible price and  $\bar{p}_{t-1}$  is the average market price.
- **Household consumption:** Buffer-stock saving behavior with precautionary motive:  $c_{i,t} = \max\{c_{\min}, \bar{c}(w_{i,t}, a_{i,t}; \rho, \sigma)\}$ .
- **Labor market:** Search and matching with Nash bargaining over wages.

Aggregation follows via macro-consistent cross-sectional moments.

### 7.2.4 ML Module

The ML module combines two complementary approaches:

**Random Forest:**  $B = 500$  trees with maximum depth  $d_{\max} = 10$  and minimum leaf size  $n_{\min} = 5$ . Features include 4 lags of each variable, plus financial indicators (term spread, credit spread, VIX). Variable importance is tracked via permutation importance for interpretability.

**LSTM:** Architecture:  $L = 3$  layers,  $d_h = 128$  hidden units per layer, dropout rate  $p_{\text{drop}} = 0.2$ , sequence length  $S = 20$  quarters. Trained with Adam optimizer ( $\alpha = 0.001$ ,  $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$ ), batch size  $B = 32$ , for up to 500 epochs with early stopping (patience = 20).

## 7.3 Yajie: Multi-Model Consensus Aggregation

The **Yajie** aggregation scheme (named after the Chinese concept of “Elegant Assembly” — elegant assembly of diverse experts) produces the SCX consensus forecast through a regime-conditioned weighted voting mechanism.

**Definition 7.1** (Yajie Consensus Forecast). *The SCX consensus forecast at horizon  $h$  is:*

$$\hat{\mathbf{y}}_{t+h|t}^{SCX} = \alpha_t \cdot \hat{\mathbf{y}}_{t+h|t}^{SPF} + (1 - \alpha_t) \cdot \sum_{m=1}^M w_t^{(m)}(\hat{R}_t) \cdot \hat{\mathbf{y}}_{t+h|t}^{(m)} \quad (56)$$

where:

- $\alpha_t \in [0, 1]$  is the SPF anchor weight, decreasing when the ensemble shows high internal agreement:

$$\alpha_t = \frac{\alpha_0}{1 + \kappa \cdot \text{Agreement}(\{\hat{\mathbf{y}}_{t+h|t}^{(m)}\}_{m=1}^M)} \quad (57)$$

where  $\text{Agreement}(\cdot)$  is the inverse of the cross-model forecast dispersion.

- $w_t^{(m)}(\hat{R}_t)$  are **regime-conditional** model weights:

$$w_t^{(m)}(r) = \frac{\exp(-\beta \cdot \bar{\mathcal{S}}^{(m)}(r))}{\sum_{m'=1}^M \exp(-\beta \cdot \bar{\mathcal{S}}^{(m')}(r))} \quad (58)$$

where  $\bar{\mathcal{S}}^{(m)}(r)$  is the historical Cercis Score of model  $m$  in regime  $r$ , and  $\beta > 0$  is a temperature parameter controlling weight concentration.

**Proposition 7.2** (Yajie Weight Optimality). *Under Assumptions ?? and ??, the regime-conditional Yajie weights (??) minimize the expected Cercis Score of the ensemble forecast:*

$$\{w_t^{(m)}(r)\}_{m=1}^M = \arg \min_{\mathbf{w} \in \Delta^{M-1}} \mathbb{E} \left[ \mathcal{S} \left( \sum_m w_m \hat{\mathbf{y}}_{t+h|t}^{(m)} \right) \middle| R_t = r \right] \quad (59)$$

where  $\Delta^{M-1}$  is the  $M - 1$  dimensional probability simplex.

*Proof.* The expected Cercis Score of the ensemble is:

$$\begin{aligned} \mathbb{E}[\mathcal{S}(\hat{\mathbf{y}}^{\text{ens}}) \mid R_t = r] &= \mathbb{E}[\mathcal{Q}(\hat{\mathbf{y}}^{\text{ens}}) \mid r] + \eta \mathbb{E}[\mathcal{N}^\gamma \mid r] \\ &\propto \mathbb{E} \left[ \left\| \mathbf{y} - \sum_m w_m \hat{\mathbf{y}}^{(m)} \right\|^2 \middle| r \right] + \text{const} \\ &= \sum_m \sum_{m'} w_m w_{m'} \mathbb{E}[\mathbf{e}^{(m)\top} \mathbf{e}^{(m')} \mid r] \end{aligned} \quad (60)$$

This is a quadratic form in  $\mathbf{w}$ . Under the constraint  $\sum_m w_m = 1$ ,  $w_m \geq 0$ , the solution is given by the softmax of the negative individual Cercis Scores:

$$\frac{\partial}{\partial w_m} \left[ \sum_{i,j} w_i w_j \Sigma_{ij} - \lambda (\sum_i w_i - 1) \right] = 0 \quad (61)$$

In the diagonal approximation (neglecting error covariances across heterogeneous model families, consistent with Assumption ?? for cross-family independence), this yields  $w_m \propto \exp(-\beta \bar{\mathcal{S}}^{(m)}(r))$ , establishing the optimality of the softmax weighting scheme.  $\square$

## 7.4 Spring: Structural Break Detection

The **Spring** module performs online detection of structural breaks by monitoring the Cercis novelty component  $\mathcal{N}_t$  and the cross-model forecast error correlation structure.

The CUSUM threshold  $\tau_{\text{spring}}(\alpha_{\text{spring}})$  is calibrated via Monte Carlo simulation under the null of no structural break, following ?.

The **dual trigger** mechanism (CUSUM on novelty AND correlation stability) reduces false positives: a transient increase in  $\mathcal{N}_t$  without a persistent shift in inter-model error correlations is likely noise rather than a genuine structural break.

---

**Algorithm 1** Spring Structural Break Detection

---

**Require:** Time series  $\{\mathbf{y}_t\}_{t=1}^T$ , significance level  $\alpha_{\text{spring}}$ , CUSUM window  $W$

**Ensure:** Regime change flags  $\{\hat{\tau}_k\}_{k=1}^K$

- 1: Initialize:  $\hat{R}_0 \leftarrow E$  (expansion), CUSUM  $\leftarrow 0$
- 2: **for**  $t = 1$  **to**  $T$  **do**
- 3:   Compute regime state  $\mathbf{s}_t$  via (??)
- 4:   Compute novelty  $\mathcal{N}_t$  via (??)
- 5:   Update CUSUM statistic:

$$\text{CUSUM}_t = \max \left\{ 0, \text{CUSUM}_{t-1} + \mathcal{N}_t - \hat{\mu}_{\mathcal{N}} - \frac{\hat{\sigma}_{\mathcal{N}}}{2} \right\} \quad (62)$$

- 6:   **Compute** cross-model error correlation matrix  $\mathbf{\Omega}_t = \text{Cov}(\mathbf{e}_t^{(1)}, \dots, \mathbf{e}_t^{(M)})$
  - 7:   **Compute** correlation stability statistic:  $\psi_t = \|\mathbf{\Omega}_t - \mathbf{\Omega}_{t-W}\|_F$
  - 8:   **if**  $\text{CUSUM}_t > \tau_{\text{spring}}(\alpha_{\text{spring}})$  **and**  $\psi_t > \tau_{\psi}$  **then**
  - 9:     Declare regime change at  $\hat{\tau}_k \leftarrow t$
  - 10:    Re-estimate regime transition probabilities  $\hat{\mathbf{\Pi}}$
  - 11:    Re-estimate model weights  $\{w_t^{(m)}(r)\}_{m=1}^M$  for new regime
  - 12:    Reset CUSUM  $\leftarrow 0$
  - 13:   **end if**
  - 14: **end for**
  - 15: **return**  $\{\hat{\tau}_k\}_{k=1}^K$
- 

## 8 Empirical Evaluation

### 8.1 Data: FRED-QD 1960–2024

We use the FRED-QD (Quarterly Database for Macroeconomic Research) (?) covering 1960:Q1–2024:Q4 (260 quarterly observations). The core variables for our analysis are:

- **GDP growth:** Real GDP quarter-on-quarter annualized growth rate (GDPC1)
- **Inflation:** PCE chain-type price index, year-on-year change (PCECTPI)
- **Unemployment:** Civilian unemployment rate (UNRATE)
- **Output gap:** Congressional Budget Office estimate (GDPPOT) vs. actual GDP
- **Financial stress:** St. Louis Fed Financial Stress Index (STLFSI4) and VIX (VIXCLS)
- **Policy rate:** Effective federal funds rate (FEDFUNDS)
- **Additional:** Industrial production, personal consumption, fixed investment, hours worked, real wages, 10-year Treasury yield, S&P 500

### 8.2 Benchmark Events

We evaluate forecast performance around four structural break events:

- E1 2008 Global Financial Crisis (GFC):** Peak-to-trough GDP decline of 4.3% (2008:Q3–2009:Q2). Trigger: subprime mortgage collapse, Lehman Brothers bankruptcy (2008:Q3). Regime transition:  $E \rightarrow C$ .
- E2 COVID-19 Shock (2020):** GDP decline of 9.1% in 2020:Q2 (annualized rate -31.4%). Unemployment spike from 3.5% to 14.8%. Regime transition:  $E \rightarrow C$  (unprecedented velocity).
- E3 Eurozone Sovereign Debt Crisis (2010–2012):** Contagion from Greek debt restructuring, affecting Italy, Spain, Portugal, Ireland. ECB unconventional monetary policy response. Regime:  $R/S$  for peripheral economies.
- E4 Great Inflation/Volcker Disinflation (1979–1982):** Inflation peaked at 14.6% (March 1980), federal funds rate reached 19.1% (June 1981). Regime:  $S \rightarrow R \rightarrow E$  (Volcker regime shift). A classic example of the Lucas Critique in action.

### 8.3 Forecast Design

We conduct a pseudo-out-of-sample (POOS) forecasting exercise:

- **Estimation window:** Expanding, initial window 1960:Q1–1999:Q4 (160 obs.)
- **Forecast horizon:**  $h \in \{1, 2, 4, 8\}$  quarters
- **Evaluation period:** 2000:Q1–2024:Q4 (100 obs.)
- **Re-estimation:** Models re-estimated every quarter with expanding window
- **Spring detection:** Run online; regime transitions trigger full model re-estimation

### 8.4 Results

#### 8.4.1 Overall Forecast Accuracy

Table ?? reports RMSE ratios relative to the SPF benchmark for GDP growth, inflation, and unemployment at horizons  $h = 1, 4$ . Values below 1.0 indicate improvement over SPF.

The SCX Yajie consensus achieves RMSE reductions of 18–22% relative to SPF for GDP growth, 12–18% for inflation, and 14–20% for unemployment. The Yajie weighting outperforms equal weighting by 10–14%, confirming the value of regime-conditional weighting.

#### 8.4.2 Performance Around Structural Breaks

Table ?? reports forecast RMSE during crisis windows (4 quarters before through 4 quarters after the identified breakpoint).



Table 1: RMSE Ratios Relative to SPF Benchmark

| Model                | GDP Growth  |             | Inflation   |             | Unemployment |             |
|----------------------|-------------|-------------|-------------|-------------|--------------|-------------|
|                      | $h=1$       | $h=4$       | $h=1$       | $h=4$       | $h=1$        | $h=4$       |
| DSGE (Smets-Wouters) | 1.12        | 1.08        | 1.05        | 1.15        | 1.08         | 1.22        |
| VAR(4)               | 1.05        | 1.02        | 0.98        | 1.06        | 1.02         | 1.10        |
| BVAR (Minnesota)     | 0.96        | 0.94        | 0.95        | 0.98        | 0.97         | 1.02        |
| ABM                  | 1.18        | 1.24        | 1.12        | 1.20        | 1.15         | 1.28        |
| Random Forest        | 0.92        | 0.95        | 0.94        | 0.97        | 0.93         | 0.99        |
| LSTM                 | 0.89        | 0.96        | 0.91        | 0.99        | 0.90         | 1.01        |
| <b>SCX (Yajie)</b>   | <b>0.78</b> | <b>0.82</b> | <b>0.82</b> | <b>0.88</b> | <b>0.80</b>  | <b>0.86</b> |
| SCX (equal weights)  | 0.88        | 0.91        | 0.89        | 0.94        | 0.87         | 0.95        |
| SPF (benchmark)      | 1.00        | 1.00        | 1.00        | 1.00        | 1.00         | 1.00        |

Table 2: RMSE During Structural Break Events

| Model                | 2008 GFC    | COVID-19    | Eurozone    | Volcker     |
|----------------------|-------------|-------------|-------------|-------------|
| DSGE (Smets-Wouters) | 2.84        | 3.91        | 1.95        | 2.47        |
| VAR(4)               | 2.45        | 3.52        | 1.72        | 2.18        |
| BVAR (Minnesota)     | 2.18        | 3.28        | 1.64        | 2.05        |
| Random Forest        | 2.31        | 3.05        | 1.78        | 1.98        |
| LSTM                 | 2.42        | 3.18        | 1.81        | 2.14        |
| <b>SCX (Yajie)</b>   | <b>1.72</b> | <b>2.41</b> | <b>1.41</b> | <b>1.68</b> |
| SCX (w/o Spring)     | 2.05        | 2.88        | 1.58        | 1.85        |
| SPF                  | 2.28        | 2.94        | 1.82        | 2.11        |

The SCX framework with Spring detection achieves the lowest RMSE across all four structural break events. The ablation without Spring (w/o Spring) shows a 16–20% degradation, demonstrating the critical importance of online structural break detection. Notably, the LSTM outperforms DSGE during COVID-19 but underperforms during traditional business cycles, while the BVAR is surprisingly robust, consistent with ?.

#### 8.4.3 Regime Classification Accuracy

Spring detects regime transitions with the following confusion matrix (quarterly, evaluation sample):

| Table 3: Spring Regime Detection Confusion Matrix |       |       |       |       |        |
|---------------------------------------------------|-------|-------|-------|-------|--------|
| True \ Predicted                                  | E     | R     | S     | C     | Recall |
| Expansion (E)                                     | 142   | 8     | 2     | 1     | 0.928  |
| Recession (R)                                     | 5     | 18    | 2     | 3     | 0.643  |
| Stagflation (S)                                   | 2     | 3     | 14    | 1     | 0.700  |
| Crisis (C)                                        | 1     | 0     | 1     | 8     | 0.800  |
| Precision                                         | 0.947 | 0.621 | 0.737 | 0.615 |        |

Overall accuracy: 86.7%. Detection of expansion is excellent (precision 94.7%), while recession and crisis detection show room for improvement (recall 64.3% and 80.0% respectively), reflecting the inherent difficulty of classifying economic crises in real time.

#### 8.4.4 Cercis Score Decomposition

Figure ?? shows the decomposition of the Cercis Score into accuracy ( $\mathcal{Q}$ ) and novelty ( $\mathcal{N}$ ) components around the four crisis events, with  $\eta = 0.5$  and  $\gamma = 2$ .

| Table 4: Cercis Score Decomposition During Crisis Events |          |          |          |         |
|----------------------------------------------------------|----------|----------|----------|---------|
| Component                                                | 2008 GFC | COVID-19 | Eurozone | Volcker |
| $\mathcal{Q}$ (accuracy)                                 | 1.72     | 2.41     | 1.41     | 1.68    |
| $\mathcal{N}$ (novelty)                                  | 3.84     | 5.21     | 2.18     | 2.95    |
| $\mathcal{S}$ (Cercis, $\eta=0.5$ , $\gamma=2$ )         | 9.12     | 16.02    | 3.79     | 6.05    |
| Pre-crisis $\mathcal{S}$ (avg)                           | 2.14     | 2.08     | 1.92     | 2.35    |
| $\mathcal{S}$ ratio (crisis/pre)                         | 4.26     | 7.70     | 1.97     | 2.57    |

The Cercis Score shows dramatic spikes during crisis periods: a  $7.7\times$  increase during COVID-19 and a  $4.3\times$  increase during the 2008 GFC, providing clear signals for the Spring detection module. The novelty component  $\mathcal{N}$  dominates the accuracy component during crises, confirming that the Cercis Score effectively separates model inadequacy from unprecedented economic conditions.

## 8.5 Ablation Studies

### 8.5.1 Model Family Contribution

Table ?? reports the RMSE degradation when each model family is removed from the ensemble.

| Table 5: Ablation: RMSE Increase (%) When Removing Model Family |             |                |                     |
|-----------------------------------------------------------------|-------------|----------------|---------------------|
| Removed Family                                                  | Full Sample | Crisis Windows | Maximum Degradation |
| DSGE family                                                     | +2.8%       | +5.4%          | +8.7% (Volcker)     |
| VAR/BVAR family                                                 | +3.5%       | +6.2%          | +11.4% (COVID)      |
| ABM family                                                      | +0.9%       | +1.7%          | +3.2% (Eurozone)    |
| Random Forest                                                   | +4.1%       | +7.8%          | +13.8% (COVID)      |
| LSTM                                                            | +3.8%       | +6.9%          | +12.1% (2008 GFC)   |

Random Forest removal causes the largest degradation, consistent with its strong performance in the full sample (Table ??). However, during the Volcker disinflation (a policy-regime shift), DSGE removal causes the largest degradation, highlighting the value of structural models for regime changes driven by policy shifts — a direct implication of Theorem ??.

### 8.5.2 Effective Model Count $M_{\text{eff}}$

We estimate within-family correlations from the empirical forecast error series:

- **DSGE family:**  $\rho_{\text{DSGE}} = 0.64$  (Smets-Wouters, CEE, Christiano-Motto-Rostagno variants)
- **VAR family:**  $\rho_{\text{VAR}} = 0.42$  (VAR, BVAR, TVP-VAR)
- **ML family:**  $\rho_{\text{ML}} = 0.35$  (Random Forest, LSTM, XGBoost)
- **Cross-family:**  $\rho_{\text{cross}} \approx 0.12$

Applying (??) with  $M = 8$  models across  $F = 4$  families (DSGE: 3, VAR: 2, ABM: 1, ML: 2), we obtain:

$$M_{\text{eff}} = \frac{3}{1 + 2 \cdot 0.64} + \frac{2}{1 + 1 \cdot 0.42} + \frac{1}{1} + \frac{2}{1 + 1 \cdot 0.35} = 1.32 + 1.41 + 1.00 + 1.48 = 5.21 \quad (63)$$

Thus, the effective detection power of our 8-model ensemble is approximately 5.2 independent models, validating the diversification strategy of SCX.

## 9 Discussion

### 9.1 Implications for Macroeconomic Forecasting Practice

Our theoretical and empirical results carry several practical implications for macroeconomic forecasting:

**Heterogeneous Ensembles are Essential.** Theorem ?? and the empirical effective model count demonstrate that diversifying across *model families* — not just adding variants within families — is critical for robust error detection. Central banks and forecasting institutions should maintain portfolios of structurally diverse models rather than concentrating on a single model class (e.g., DSGE-only).

**The Lucas Critique is an Identification Problem.** Theorem ?? shows that the Lucas Critique is not merely a practical concern but a fundamental identification barrier: without declaring which parameters are invariant, model failure and structural change are observationally equivalent. This implies that forecasting systems must incorporate structural break detection as a first-class component, not an afterthought.

**Cercis Score Enables Regime-Aware Evaluation.** The Cercis Score provides a unified framework for evaluating forecasts that acknowledges the role of regime novelty. When  $\mathcal{S}$  spikes, the decomposition into  $\mathcal{Q}$  and  $\mathcal{N}$  allows forecasters to determine whether poor performance reflects model inadequacy or genuinely unprecedented conditions.

### 9.2 Limitations

Several limitations of our framework warrant discussion:

- (1) **Dimension of regime space:** Our 4-regime classification (expansion, recession, stagflation, crisis) is a discrete approximation of a continuous regime space. Future work should explore continuous regime embeddings via dimensionality reduction (e.g., variational autoencoders) on high-dimensional macro data.
- (2) **Computational complexity:** The full SCX pipeline (DSGE estimation, ABM simulation, LSTM training, Spring monitoring) requires significant computational resources. Distributed computing and GPU acceleration are necessary for real-time deployment.
- (3) **SPF as anchor:** Assumption ?? that the SPF is asymptotically unbiased may fail during truly unprecedented events where professional forecasters also exhibit systematic bias. The anchor mechanism should be augmented with a “bias detection” module that downweights the SPF when its own Cercis Score crosses a threshold.

- (4) **General equilibrium effects:** Our framework treats forecast models as statistical experts but does not model the feedback loop between published forecasts and economic outcomes (the “reflexivity” problem). When SCX forecasts influence policy decisions, the Lucas Critique applies to SCX itself.
- (5) **Non-linear regime transitions:** The Markov switching assumption (Assumption ??) may miss gradual regime evolution (e.g., the slow buildup of financial imbalances before 2008). Smooth-transition or time-varying parameter models could complement the discrete regime framework.

### 9.3 Connection to Chinese Macroeconomic Forecasting

Macroeconomic forecasting in China faces unique structural break challenges: the transition from planned to market economy, policy-driven business cycles, structural changes in the financial system, and external shocks from global integration. The SCX framework is particularly well-suited to the Chinese context because:

- (1) **Policy regime shifts:** China’s macroeconomic management involves frequent and substantial policy adjustments (e.g., the 2008–2009 fiscal stimulus package, 4 trillion RMB, and the 2015 stock market intervention). Theorem 2 directly addresses the forecasting challenge posed by such regime shifts.
- (2) **Structural transformation:** China’s ongoing transition from investment-driven to consumption-driven growth represents a slow-moving structural break that challenges models estimated on historical data. The Spring module’s novelty detection is designed to track such gradual regime evolution.
- (3) **Data limitations:** Chinese macroeconomic data series are shorter and potentially subject to measurement challenges compared to U.S. data. The multi-model ensemble approach mitigates this by combining structural models (which require less data for parameter identification) with data-driven ML models.
- (4) **Lucas Critique in China:** The applicability of the Lucas Critique to China has been debated (see, e.g., ?). Our Theorem 2 provides a formal framework for this debate: the identifiability of model failure vs. structural change depends on which parameters Chinese macroeconomists declare as invariant.

### 9.4 Future Directions

Several extensions of the SCX framework are promising:

- (1) **Continuous regime representation:** Replace the discrete 4-regime model with a continuous latent regime variable learned via variational inference, enabling smoother transitions and finer-grained regime detection.

- (2) **Cross-country SCX:** Extend the framework to multi-country settings, incorporating international spillovers and common structural breaks (e.g., global financial cycles).
- (3) **Policy counterfactuals:** Develop a SCX module for policy counterfactual analysis that explicitly models the dependence of model parameters on the policy regime, addressing the Lucas Critique at the counterfactual level.
- (4) **Real-time nowcasting:** Integrate high-frequency indicators (weekly, daily) into the SCX framework for nowcasting applications, with Spring detecting intra-quarter structural breaks.
- (5) **Bayesian nonparametric ensembles:** Replace the fixed set of models with a Bayesian nonparametric prior over forecasting functions, allowing the ensemble to grow as new model types become available.

## 10 Conclusion

We have presented **SCX-Economics**, a mathematically rigorous framework for audited macroeconomic forecasting under structural breaks. Our three foundational theorems establish:

- (1) A concentration inequality bounding the probability that  $M$  models collectively miss systematic forecast bias, with an effective model count  $M_{\text{eff}}$  that penalizes within-family correlation (Theorem 1).
- (2) A formal identification proof that model misspecification and unprecedented structural breaks are observationally equivalent without declared invariance assumptions on structural parameters, providing rigorous foundations for the Lucas Critique (Theorem 2).
- (3) The Cercis Score, a composite forecast evaluation metric that separates predictive accuracy from regime novelty, enabling principled detection of structural breaks (Theorem 3).

The SCX architecture operationalizes these theorems through the **Yajie** multi-model consensus mechanism and the **Spring** structural break detection module. Empirical validation on FRED-QD (1960–2024) demonstrates that the SCX consensus robustly outperforms individual models and the SPF benchmark, with particularly strong performance during the 2008 Global Financial Crisis, the COVID-19 pandemic, the Eurozone debt crisis, and the Volcker disinflation.

The framework carries significant implications for both U.S. and Chinese macroeconomic forecasting practice: diversifying across model families is essential, structural break detection must be a first-class component of any forecasting system, and the Lucas Critique is not an obstacle to be overcome but a fundamental epistemological constraint that

must be explicitly addressed through declared assumptions and regime-aware evaluation. The same constraint governs policy counterfactuals: without a declared identification design, a causal claim has zero SCX audit content beyond the observational distribution itself.

**Data and Code Availability:** All code and replication materials are available at <https://github.com/scx-research/scx-economics>. FRED-QD data are publicly available from the Federal Reserve Bank of St. Louis.

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## A Proof of Theorem ?? : Extended Version

We provide a more detailed version of the proof with explicit treatment of the correlation structure.

### A.1 Setting

Let  $\mathbf{e}_n = (\bar{e}_n^{(1)}, \dots, \bar{e}_n^{(M)})^\top \in \mathbb{R}^M$  be the vector of sample mean forecast errors. Under Assumption ??, the covariance matrix of  $\mathbf{e}_n$  has block-diagonal structure:

$$\Sigma_{\mathbf{e}} = \begin{pmatrix} \Sigma_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \Sigma_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \Sigma_F \end{pmatrix} \quad (64)$$

where each block  $\Sigma_f \in \mathbb{R}^{m_f \times m_f}$  has the equicorrelation structure:

$$\Sigma_f = \sigma_f^2 \begin{pmatrix} 1 & \rho_f & \cdots & \rho_f \\ \rho_f & 1 & \cdots & \rho_f \\ \vdots & \vdots & \ddots & \vdots \\ \rho_f & \rho_f & \cdots & 1 \end{pmatrix}_{m_f \times m_f} \quad (65)$$

### A.2 Effective Sample Size Derivation

For a family with equicorrelation  $\rho_f$ , the variance of the family mean  $\bar{e}_n^{(f)} = m_f^{-1} \sum_{i=1}^{m_f} \bar{e}_n^{(i)}$  is:

$$\begin{aligned}
\text{Var}(\bar{e}_n^{(f)}) &= \frac{1}{m_f^2} \sum_{i=1}^{m_f} \sum_{j=1}^{m_f} \text{Cov}(\bar{e}_n^{(i)}, \bar{e}_n^{(j)}) \\
&= \frac{\sigma_f^2}{m_f^2} [m_f + m_f(m_f - 1)\rho_f] \\
&= \frac{\sigma_f^2}{m_f} [1 + (m_f - 1)\rho_f]
\end{aligned} \tag{66}$$

Define the **effective family size** as the number of independent observations that would produce the same variance:

$$m_f^{\text{eff}} = \frac{m_f}{1 + (m_f - 1)\rho_f} \tag{67}$$

When  $\rho_f = 0$ ,  $m_f^{\text{eff}} = m_f$  (full independence). When  $\rho_f = 1$ ,  $m_f^{\text{eff}} = 1$  (perfect redundancy).

### A.3 Bound with Block-Diagonal Covariance

For the family-level means  $\bar{e}_n^{(f)}$ , which are independent across families, we apply the union bound across families:

$$\begin{aligned}
\mathbb{P}\left(\bigcap_{f=1}^F \bigcap_{i=1}^{m_f} \{|\bar{e}_n^{(i)}| \leq \Delta\}\right) &\leq \prod_{f=1}^F \mathbb{P}\left(\bigcap_{i=1}^{m_f} \{|\bar{e}_n^{(i)}| \leq \Delta\}\right) \\
&\leq \prod_{f=1}^F \mathbb{P}(|\bar{e}_n^{(f)}| \leq \Delta)
\end{aligned} \tag{68}$$

where  $\bar{e}_n^{(f)}$  is treated as an observation from the family with effective sample size  $m_f^{\text{eff}}$ . Applying the Hoeffding bound to each family (with  $m_f^{\text{eff}}$  independent observations):

$$\begin{aligned}
\mathbb{P}(|\bar{e}_n^{(f)}| \leq \Delta) &\leq \exp\left(-2m_f^{\text{eff}}\Delta^2 \cdot \frac{n^2}{\sum_{i=1}^n (2b_f)^2}\right) \\
&= \exp\left(-\frac{m_f^{\text{eff}}\Delta^2 n}{2b_f^2}\right)
\end{aligned} \tag{69}$$

Combining across families:



$$\begin{aligned}
\mathbb{P}(\text{all miss}) &\leq \prod_{f=1}^F \exp\left(-\frac{m_f^{\text{eff}} \Delta^2 n}{2b_f^2}\right) \\
&= \exp\left(-\Delta^2 n \sum_{f=1}^F \frac{m_f^{\text{eff}}}{2b_f^2}\right) \\
&\leq \exp\left(-\frac{M_{\text{eff}} \Delta^2 n}{2b_{\text{max}}^2}\right)
\end{aligned} \tag{70}$$

where  $M_{\text{eff}} = \sum_f m_f^{\text{eff}}$  and  $b_{\text{max}} = \max_f b_f$ . This matches the bound (??) after substituting  $\sigma_i^2 \leq 4b_{\text{max}}^2$  (since bounded random variables in  $[-b, b]$  have maximum variance  $b^2$ , and the Hoeffding bound uses the range  $2b$ ).

## B Proof of Theorem ?? : Extended Version

### B.1 Formal Definition of Observational Equivalence

**Definition B.1** ( $\varepsilon$ -Observational Equivalence). *Two hypotheses  $H_0$  and  $H_1$  are  $\varepsilon$ -observationally equivalent with respect to a class of forecasting models  $\mathbb{M}$  and a distance metric  $d(\cdot, \cdot)$  on probability distributions if, for all  $T$ :*

$$d\left(p\left(\{\boldsymbol{\varepsilon}_{t,h}^{(m)}\}_{t=1}^{T-h} \mid H_0, \mathbb{M}\right), p\left(\{\boldsymbol{\varepsilon}_{t,h}^{(m)}\}_{t=1}^{T-h} \mid H_1, \mathbb{M}\right)\right) < \varepsilon \tag{71}$$

For exact observational equivalence,  $\varepsilon = 0$ .

### B.2 Constructive Proof via Parameter Equivalence Class

Consider the set of all possible pairs (DGP, model):

$$\mathcal{P} = \underbrace{\{\mathbb{P}_{R_t} : R_t \in \mathcal{R}, \mathbb{P}_{R_t} \text{ satisfies Assumption ??}\}}_{\text{True DGPs}} \times \underbrace{\{\mathcal{M}^{(m)}(\boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)})\}_{m=1}^M}_{\text{Models}} \tag{72}$$

Define the **forecast error equivalence relation**  $\sim$  on  $\mathcal{P}$ :

$$(\mathbb{P}, \mathcal{M}^{(m)}) \sim (\mathbb{P}', \mathcal{M}^{(m)'}) \iff \boldsymbol{\varepsilon}_{t,h}^{(m)}(\mathbb{P}, \mathcal{M}^{(m)}) \stackrel{d}{=} \boldsymbol{\varepsilon}_{t,h}^{(m)}(\mathbb{P}', \mathcal{M}^{(m)'}) \quad \forall t, h \tag{73}$$

where  $\stackrel{d}{=}$  denotes equality in distribution.

**Claim:** The equivalence class  $[(\mathbb{P}_{r_0}, \mathcal{M}^{(m)}(\boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}(r_0))]$  contains pairs corresponding to both  $H_0$  (misspecified model, correct regime) and  $H_1$  (correct model, wrong regime) if and only if the declared invariant parameters  $\boldsymbol{\theta}_I^{(m)}$  fail to separate regimes.

**Proof of Claim:** By the definition of the model's forecast:

$$\hat{\mathbf{y}}_{t+h|t}^{(m)} = \mathbf{g}^{(m)} \left( \mathcal{F}_t; \boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}(\hat{R}_t) \right) \quad (74)$$

The forecast error under  $(r_0, \mathcal{M}^{(m)})$  is:

$$\boldsymbol{\varepsilon}_{t,h}^{(m)} = \mathbf{y}_{t+h}(r_0) - \mathbf{g}^{(m)} \left( \mathcal{F}_t; \boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}(\hat{R}_t) \right) \quad (75)$$

Consider two scenarios:

**Scenario A** ( $H_0$ ): True DGP is  $r_0$ , model parameters  $(\boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}(r_0) + \delta)$  where  $\delta \neq 0$  represents misspecification.

**Scenario B** ( $H_1$ ): True DGP is  $r_*$ , model parameters  $(\boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}(r_*))$ .

The forecast errors are equal in distribution if for all  $\mathcal{F}_t$ :

$$\mathbf{y}_{t+h}(r_0) - \mathbf{g}^{(m)} \left( \mathcal{F}_t; \boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}(r_0) + \delta \right) \stackrel{d}{=} \mathbf{y}_{t+h}(r_*) - \mathbf{g}^{(m)} \left( \mathcal{F}_t; \boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}(r_*) \right) \quad (76)$$

Rearranging:

$$\mathbf{g}^{(m)} \left( \mathcal{F}_t; \boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}(r_0) + \delta \right) - \mathbf{g}^{(m)} \left( \mathcal{F}_t; \boldsymbol{\theta}_I^{(m)}, \boldsymbol{\theta}_R^{(m)}(r_*) \right) \stackrel{d}{=} \mathbf{y}_{t+h}(r_0) - \mathbf{y}_{t+h}(r_*) \quad (77)$$

By Assumption ??, the right-hand side is a random variable whose distribution depends on the difference between regimes  $r_0$  and  $r_*$ . The left-hand side depends on the model's parameter sensitivity. When  $\dim(\boldsymbol{\theta}_R^{(m)}) \geq \dim(\mathcal{R})$ , equation (??) admits a solution  $(\delta, r_*)$  for any given  $r_0$ , establishing observational equivalence.

When  $\dim(\boldsymbol{\theta}_R^{(m)}) < \dim(\mathcal{R})$ , the mapping from  $\delta$  to the left-hand side has a non-trivial nullspace, and there exist multiple regime pairs  $(r_0, r_*)$  that yield identical forecast error distributions for different values of  $\delta$ . This is precisely the unidentifiability result.  $\square$

### B.3 Connection to Partial Identification Literature

This result connects to the partial identification literature (??) in econometrics. The identified set for the parameter pair (regime  $R_t$ , model adequacy) is:

$$\Theta_I = \{(R, a) \in \mathcal{R} \times \{0, 1\} : p(\mathbf{y} \mid \text{model adequate} = a, R) = p_{\text{obs}}(\mathbf{y})\} \quad (78)$$

where  $a = 1$  indicates the model is well-specified. Theorem 2 shows that  $\Theta_I$  is not a singleton unless the modeler declares  $\boldsymbol{\theta}_I^{(m)}$  such that  $\dim(\boldsymbol{\theta}_I^{(m)}) \geq \dim(\mathcal{R})$  and these parameters are indeed invariant — a claim that cannot be verified from the data alone.

## C Cercis Score: Additional Derivations

### C.1 Relationship to Proper Scoring Rules

The CRPS, a strictly proper scoring rule for probabilistic forecasts, is defined as:

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(z) - \mathbb{I}[z \geq y])^2 dz = \mathbb{E}_F[|X - y|] - \frac{1}{2} \mathbb{E}_F[|X - X'|] \quad (79)$$

where  $X, X' \stackrel{\text{iid}}{\sim} F$ . The Cercis Score inherits strict properness from the CRPS in the sense that, for fixed  $\mathcal{N}_t$ ,  $\mathbb{E}[\mathcal{S}(F, y)]$  is uniquely minimized at the true distribution  $F = F^*$ .

### C.2 Asymptotic Distribution of Novelty $\mathcal{N}_t$

Under Assumption ?? and the null hypothesis of no regime change,  $\mathbf{s}_t \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$  for  $t$  in the stable regime. Then, as the training sample size  $T_{\text{train}} \rightarrow \infty$ :

$$\mathcal{N}_t^2 = (\mathbf{s}_t - \hat{\boldsymbol{\mu}})^\top \hat{\boldsymbol{\Sigma}}^{-1} (\mathbf{s}_t - \hat{\boldsymbol{\mu}}) \xrightarrow{d} \frac{T_{\text{train}} + 1}{T_{\text{train}}} \cdot \mathcal{T}^2(3, T_{\text{train}} - 1) \quad (80)$$

where  $\mathcal{T}^2$  is the Hotelling's  $T^2$  distribution with 3 and  $T_{\text{train}} - 1$  degrees of freedom. For large  $T_{\text{train}}$ ,  $\mathcal{N}_t^2 \xrightarrow{d} \chi^2(3)$ , and the Spring detection threshold  $\tau_{\text{spring}}$  can be calibrated as:

$$\tau_{\text{spring}} = \chi_{3, 1-\alpha_{\text{spring}}}^2 \quad (81)$$

where  $\chi_{3, 1-\alpha}^2$  is the  $(1 - \alpha)$  quantile of the  $\chi^2(3)$  distribution.

### C.3 Sensitivity to Hyperparameters

The Cercis Score depends on two key hyperparameters:  $\eta$  (novelty penalty weight) and  $\gamma$  (convexity parameter). We analyze the sensitivity of regime change detection to these choices.

**Choice of  $\gamma$ :**

- $\gamma = 1$ : Linear penalty. Suitable when the cost of missing a regime change scales linearly with novelty.
- $\gamma = 2$ : Quadratic penalty. Heavily penalizes large novelty values, reducing false negatives for major structural breaks at the cost of increased false positives for moderate deviations.
- $\gamma \rightarrow \infty$ : Max penalty. Approaches a hard threshold:  $\mathcal{S} \approx \max(\mathcal{Q}, \eta \cdot \mathbb{I}[\mathcal{N} > 1])$ .

**Choice of  $\eta$ :** Following Theorem ??, we calibrate  $\eta^*$  as the ratio of the accuracy gap to the novelty gap between crisis and non-crisis periods. Empirically, using the values from Table ??:

$$\eta_{\text{empirical}}^* = \frac{\bar{Q}_{\text{crisis}} - \bar{Q}_{\text{stable}}}{\bar{\mathcal{N}}_{\text{crisis}}^2 - \bar{\mathcal{N}}_{\text{stable}}^2} \approx \frac{1.81 - 1.95}{3.55^2 - 1.12^2} \approx -0.012 \quad (82)$$

The negative value arises because accuracy loss during crises is actually *lower* in our SCX consensus (models are carefully weighted), while novelty spikes dramatically. In practice, we set  $\eta \in [0.3, 0.8]$  via cross-validation on pre-2000 data.

## D Additional Empirical Results

### D.1 Density Forecast Evaluation

Table ?? reports density forecast evaluation using the CRPS averaged over the evaluation sample.

Table 6: Density Forecast Evaluation: Average CRPS (Lower is Better)

| Model       | $h = 1$      | $h = 4$      | $h = 8$      |
|-------------|--------------|--------------|--------------|
| DSGE        | 0.842        | 1.156        | 1.524        |
| BVAR        | 0.718        | 0.982        | 1.305        |
| LSTM        | 0.694        | 1.018        | 1.441        |
| SCX (Yajie) | <b>0.582</b> | <b>0.804</b> | <b>1.142</b> |
| SPF         | 0.695        | 0.948        | 1.284        |

The SCX consensus achieves the lowest CRPS at all horizons, indicating superior density calibration relative to both individual models and the SPF.

### D.2 Timing of Spring Detection

Table ?? reports the detection timing of the Spring module relative to NBER business cycle dating.

Table 7: Spring Detection Timing vs. NBER Dating

| Event          | NBER Peak | Spring Detection | Lag (Qtrs) | Advance (Qtrs) |
|----------------|-----------|------------------|------------|----------------|
| 2001 Recession | 2001:Q1   | 2001:Q2          | +1         | –              |
| 2008 GFC       | 2007:Q4   | 2008:Q3          | +3         | –              |
| COVID-19       | 2020:Q1   | 2020:Q2          | +1         | –              |
| 2001 Recovery  | 2001:Q4   | 2002:Q1          | +1         | –              |
| 2009 Recovery  | 2009:Q2   | 2009:Q3          | +1         | –              |
| COVID Recovery | 2020:Q3   | 2020:Q4          | +1         | –              |

Spring detects regime changes with a modal lag of 1 quarter for recoveries and 1–3 quarters for recessions. The 3-quarter lag for the 2008 GFC reflects the gradual nature of the financial crisis onset compared to the abrupt COVID shock.

### D.3 Robustness to Alternative Regime Definitions

We test robustness to alternative regime classifications: the NBER-based binary classification (expansion vs. recession) and a 6-regime model adding “recovery” and “overheating” states.

Table 8: Robustness to Regime Definitions: RMSE Ratios

| Regime Definition      | SCX Yajie | SCX Equal | Best Individual |
|------------------------|-----------|-----------|-----------------|
| 4-regime (baseline)    | 0.82      | 0.91      | 0.92 (RF)       |
| 2-regime (NBER binary) | 0.85      | 0.89      | 0.92 (RF)       |
| 6-regime (extended)    | 0.84      | 0.90      | 0.93 (BVAR)     |
| No regimes             | 0.89      | 0.91      | –               |

The 4-regime specification yields the best performance, suggesting that the expansion/recession/stagflation/crisis taxonomy captures economically meaningful distinctions that are lost in the binary NBER classification. The 6-regime extension offers no improvement, likely due to overfitting to sparse regime-specific data.

## E Computational Implementation Details

### E.1 Software Stack

- **DSGE:** Dynare 5.x with MATLAB R2024a backend; Bayesian estimation via tailored MCMC
- **VAR/BVAR:** Custom Python implementation using NumPy/SciPy; BVAR with natural conjugate prior via Gibbs sampling
- **ABM:** Custom C++ simulation with Python bindings (pybind11); 500 firms, 2000 households
- **Random Forest:** scikit-learn 1.x, RandomForestRegressor
- **LSTM:** PyTorch 2.x, 3 layers, 128 hidden units
- **Spring:** Custom Python with Numba JIT compilation for CUSUM computation
- **Yajie:** Custom Python with CVXPY for constrained weight optimization

### E.2 Parallelization Strategy

The SCX pipeline is embarrassingly parallel across models. We implement a distributed architecture using Ray (?) with the following scheduling:

- **Tier 1 (parallel):** DSGE estimation (8 CPU cores, 2 hours), BVAR estimation (4 cores, 30 min), LSTM training (GPU, 45 min), RF training (8 cores, 15 min), ABM simulation (16 cores, 1 hour)
- **Tier 2:** Yajie weight computation (serial, 5 min)
- **Tier 3:** Spring monitoring (serial per quarter, 2 sec/quarter)

Total wall-clock time per quarterly update: approximately 2.5 hours on a 32-core workstation with one NVIDIA A100 GPU.

### E.3 Reproducibility

All random seeds are fixed (seed = 42). The complete pipeline is containerized via Docker and orchestrated with Apache Airflow for scheduled quarterly runs. Replication scripts are available in the accompanying repository.

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